SEMICONDUCTOR DESIGN

RISC16 SRAM Processor

RTL Verification · Multi-Process Synthesis · Challenge

16-bit RISC + dual SRAM. RTL 검증부터 SAED32 · TSMC 180 / 130 nm 합성, Challenge와 공정별 비교까지.

정태웅 · 청주대학교 · Design Compiler · Formality · VCS

Agenda

RTL 검증 → 자동화 TCL → 공정별 합성 · challenge → 최종 비교.

01	**RTL & Verification**	구조 · instruction · testbench · RTL VCS.
02	**Custom Automation TCL**	split_fanout · report_transition_drivers.
03	**Synthesis @ 2.0 ns**	SAED32 기준 — compile0 이후 위반을 수동 해결.
04	**Challenge @ 1.8 ns**	1.8 ns 수렴 vs 1.7 ns 미수렴.
05	**TSMC 180 nm**	6.5 ns 수렴 vs 6.4 ns 미수렴.
06	**TSMC 130 nm**	2.6 ns 수렴 vs 2.5 ns 미수렴 (WLM).
07	**Comparison & Wrap-Up**	공정 · topo/WLM 차이와 총정리.

Flow

모든 공정 · 주기에 동일하게 적용한 RTL-to-netlist 흐름.

RTL Simulation → **Synthesis (DC)** → **Timing / DRC Fix** → **Formality (FM)** → **Gate Netlist Sim**

RTL & Analysis

RISC16 계층 해석, SRAM preload, done/pass end point 확인.

Constrained Synthesis

spec 제약 하에 compile_ultra 매핑 · 최적화, 위반 fix.

Equivalence & Sim

Formality로 RTL ≡ netlist, gate-level VCS로 done/pass 재현(SAED32).

01

SECTION

RTL & Verification

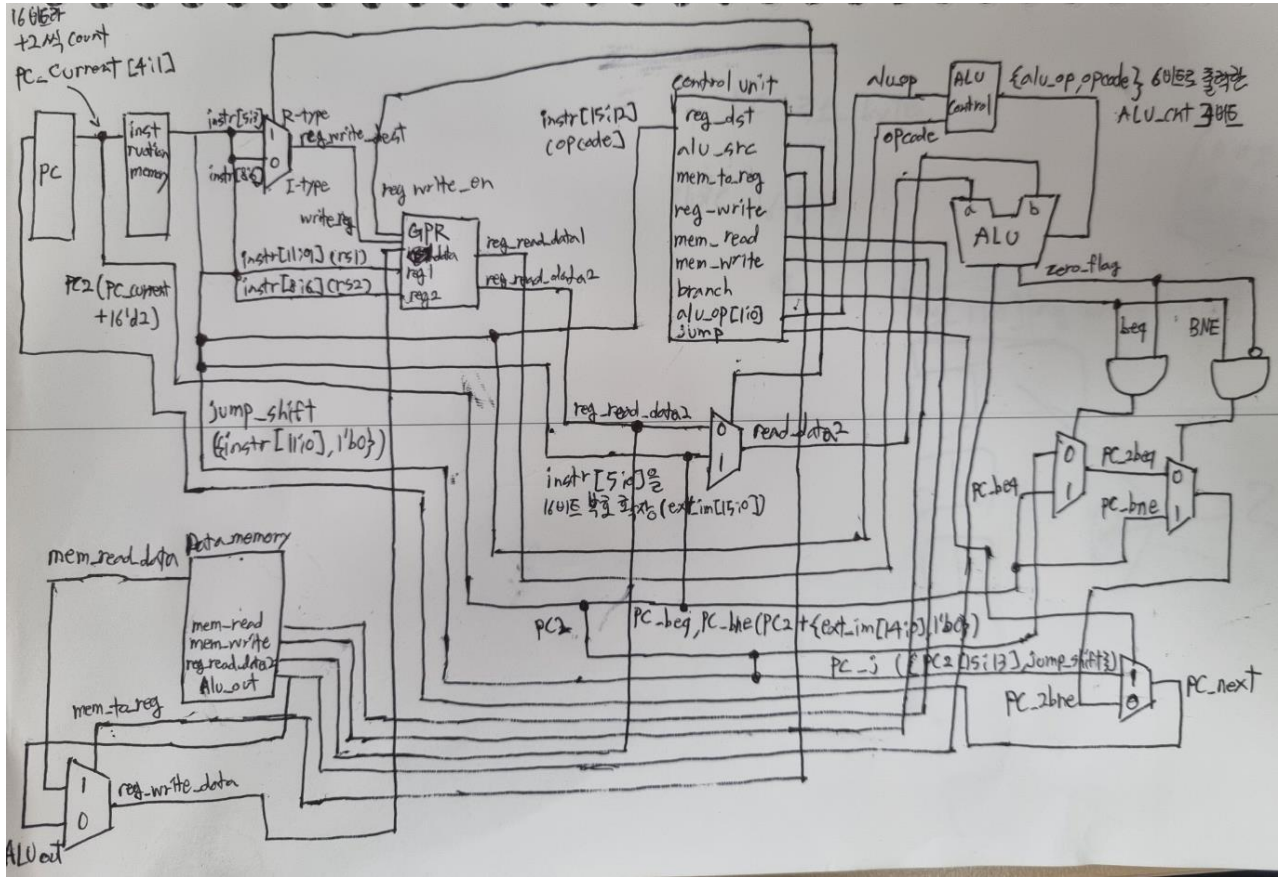
RISC16 구조와 dual-SRAM datapath, testbench와 RTL VCS로 기능 검증.

정태웅 · 청주대학교

RISC16 Architecture — Top Level(Datapath 계층 확장)

16-bit RISC: control_unit·datapath·(instruction/data)_memory·GPRs·ALU ·alu_control.

- RISC Architecture



Risc_16_bit

Control_Unit & Datapath
done,pass로 통과 판별
Parameter.v : 전역 변수 선언

Control_Unit

Opcode ->
reg_dst/alu_src/mem_to_reg/reg_write/mem_read/mem_write/branch(beq,bne)/alu_op[1:0]/jump 플래그 활성화

Datapath

byte 기준 pc word 기준 count를 위한 +2
done, pass 활성화 시, pc count stop
Instruction_memory : pc_current 입력 따라 해당 주소 instruction 출력
GPRs : register file 8*16
ALU_Control : {alu_op,opcode}로 alu_control 출력
ALU : alu_control -> ADD,SUB,NOT,SLL,SRL,AND,OR,SLT 수행.
zero_flag 출력
Data_memory : SRAM1RW64x8 ×2, 병렬 SRAM(high,low)

Instruction Format

16-bit 명령의 필드 구성 — R-type과 I-type.

instruction format

비트	[15:12]	[11:9]	[8:6]	[5:3]	[5:0]
R-type	opcode	rs1	rs2	rd	-
I-type	opcode	rs1(base)	rd	-	immediate

공통

opcode는 instr[15:12], rs1은 instr[11:9].

R-type

rs2 instr[8:6], rd instr[5:3].

I-type

rd instr[8:6], immediate instr[5:0]를 부호 확장해 사용.

Data Memory — Dual SRAM

SRAM1RW64x8 2개를 병렬 배치해 16-bit-wide data memory 구성.

SRAM1RW64x8 · pin 정의

Table 1.6. Single port SRAMpRWnxw_1rw Pin Definition

Pin Symbol	Width (bits)	Type	Name and Function
A	k	Input	Primary Read/Write Address
CE	1	Input	Primary Positive-Edge Clock
WEB	1	Input	Primary Write Enable, Active Low
OEB	1	Input	Primary Output Enable, Active Low
CSB	1	Input	Primary Chip Select, Active Low
I	n	Input	Primary Input data bus
O	n	Output	Primary Output data bus
VDD	Power supply		
VSS	Power ground		

구성

u_sram_low = data[7:0],
u_sram_high = data[15:8].

pin

A / CE / CSB / WEB / OEB / I / O.

clocking

sram_clk = ~clk → CE falling edge, half-cycle 경로.

Data Memory — pin 연결

두 macro에 같은 주소 · 제어를 주고 데이터만 상 · 하위로 분할.

Data_Memory.v · pin 연결

```
SRAM1RW64x8 u_sram_low ( // 하위 비트
    .A    (sram_addr), // address, 어느 word 읽거나 쓸지 지정하는 핀
    .CE   (sram_clk), // clock, falling edge
    .WEB  (sram_web), // Write Enable Bar, 0이면 쓰기 모드, 1이면 읽기 모드를 선택하는 핀
    .OEB  (sram_oeb), // Output Enable Bar, 0이면 O 핀에 실제 데이터 출력, 1이면 출력을 High-Z
    .CSB  (sram_csb), // Chip Select Bar, 이 SRAM 칩 자체를 활성화할지 말지 결정하는 핀. 0이면
칩이 동작하고, 1이면 이 SRAM은 아예 무시(비활성) 상태
    .I    (mem_write_data[7:0]), // input
    .O    (sram_dout_low) // output
);
```

주소 / 제어

sram_addr · sram_web · sram_oeb · sram_csb를 두 macro가 공유해 동시 access.

데이터 분할

I는 mem_write_data[7:0] / [15:8], O는 sram_dout_low / high로 16-bit 구성.

RTL — Pass 조건

done/pass 판정 로직: pc = 0x14 도달 + opcode 9 + pc_next 0x16.

RISC16 RTL · done/pass 판정

```
always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        done_r <= 1'b0;
        pass_r <= 1'b0;
    end else begin
        if (!done_r && (pc_current == 16'h0014)) begin
            done_r <= 1'b1;

            // opcode 9 instruction까지 정상 도달했고,
            // 다음 PC가 16'h0016으로 계산되면 PASS
            pass_r <= (opcode == 4'h9) &&
                      (instr  == 16'h9050) &&
                      (pc_next == 16'h0016);
        end
    end
end
```

done

!done_r && pc_current == 16'h0014 → done_r = 1.

pass

opcode == 4'h9 && instr == 16'h9050 && pc_next == 16'h0016 → pass_r = 1.

의미

deterministic end point라 이후 명령의 X/Z가 판정에 영향 없음.

Instruction Set & Test Program

ROM에 적재된 명령 시퀀스가 reset → fetch → execute를 구동.

test 프로그램 · opcode 순서

```
0000_0100_0000_0000 // load R0 <- Mem(R2 + 0)
0000_0100_0100_0001 // load R1 <- Mem(R2 + 1)
0010_0000_0101_0000 // Add R2 <- R0 + R1
0001_0010_1000_0000 // Store Mem(R1 + 0) <- R2
0011_0000_0101_0000 // sub R2 <- R0 - R1
0100_0000_0101_0000 // invert R2 <- !R0
0101_0000_0101_0000 // logical shift left R2 <- R0<<R1
0110_0000_0101_0000 // logical shift right R2 <- R0>>R1
0111_0000_0101_0000 // AND R2<- R0 AND R1
1000_0000_0101_0000 // OR R2<- R0 OR R1
1001_0000_0101_0000 // SLT R2 <- 1 if R0 < R1
0010_0000_0000_0000 // Add R0 <- R0 + R0
1011_0000_0100_0001 // BEQ branch to jump if R0=R1, PCnew= PC+2+offset<<1 = 28 => offset = 1
1100_0000_0100_0000 // BNE branch to jump if R0!=R1, PCnew= PC+2+offset<<1 = 28 => offset = 0
1101_0000_0000_0000 // J jump to the beginning address
```

명령 흐름

LW · LW · ADD · SW · SUB · NOT · SLL · SRL · AND · OR · SLT · ADD · BEQ · BNE · J.

end point

opcode 9(SLT, instr 0x9050)에서 pc = 0x14 도달 → done/pass 판정 지점. 이후 pc = 0x16에서 반복.

Testbench

reset → SRAM preload → instruction execution → done까지 자동 구동.

risc_testbench.v

```
module risc_testbench;
    reg clk;
    reg rst_n;
    wire pass, done;

    Risc_16_bit dut (
        .clk   (clk),
        .rst_n (rst_n),
        .pass  (pass),
        .done  (done)
    );

    parameter CLK_PERIOD = 20;
    initial clk = 1'b0;
    always #(CLK_PERIOD/2) clk = ~clk;

    reg [15:0] init_data [0:7];

    task preload_data_memory;
        integer a;
        begin
            for (a = 0; a < 8; a = a + 1) begin
                force dut.DU.dm.mem_access_addr = a[15:0];
                force dut.DU.dm.mem_write_data  = init_data[a];
                force dut.DU.dm.mem_write_en    = 1'b1;
                force dut.DU.dm.mem_read        = 1'b0;
                @(negedge clk);
                @(posedge clk);
            end
            release dut.DU.dm.mem_access_addr;
            release dut.DU.dm.mem_write_data;
            release dut.DU.dm.mem_write_en;
            release dut.DU.dm.mem_read;
        end
    endtask

    initial begin
        $readmemb("./test/test.data", init_data, 0, 7);

        rst_n = 1'b0;
        preload_data_memory();

        @(negedge clk);
        rst_n = 1'b1;

        while (!done) @(negedge clk);
        #(CLK_PERIOD*4);
        $finish;
    end

endmodule
```

preload

preload_data_memory에서 force로 주소 · 데이터를 주입하고 $readmemb으로 적재.

execution

rst_n 인가 후 clk 토글, while(!done)으로 명령 시퀀스 수행.

종료

done = 1 이후 일정 시간 뒤 $finish — 자동 종료.

RTL VCS — Functional Simulation

behavioral simulation으로 done/pass end point 확인.

- RTL waveform · done/pass · PC/opcode · SRAM pins



02

SECTION

Custom Automation TCL

반복되는 fanout · transition 위반 수정을 TCL proc로 자동화.

정태웅 · 청주대학교

왜 자동화했나

2.0 ns까지는 수동으로 충분했지만 1.8 ns · TSMC부터 위반 수가 급증.

문제

주기 단축과 공정 이식 시
max_fanout · max_transition 위반이
수십 개 단위로 발생.

수동의 한계

net마다 report_net 확인 후
insert_buffer / size_cell 반복 —
느리고 누락 위험.

해결

위반 net을 스스로 찾아
buffer 삽입 · driver 교체하는
TCL proc 작성 → 1.8 ns부터 적용.

두 proc 모두 공정에 종속되지 않도록 max_fanout과 buffer/driver cell을 인자로 받음 — SAED32는 NBUFFX2_LVT에 max_fanout 8, TSMC 180은 tcb018g3d3wc/BUFFD1에 max_fanout 10, TSMC 130은 scx2_tsmc_cl013g/BUFX2에 max_fanout 8을 사용.

split_fanout (1/3)

parse_report_net_loads

```
proc parse_report_net_loads {net_name} {
    redirect -variable rpt_net {report_net -connections -verbose $net_name}

    set raw_lines [split $rpt_net "\n"]

    set in_loads 0
    set pending ""
    set collected {}

    foreach raw $raw_lines {
        set trimmed [string trim $raw]
        if {$trimmed == ""} { continue }

        if {[string match "Load Pins*" $trimmed]} {
            set in_loads 1
            continue
        }
        if {!$in_loads} { continue }

        if {[string match "----*" $trimmed]} { continue }

        if {[string match "*Input Pin*" $trimmed] || [string match "*Output Pin*" $trimmed]} {
            if {$pending != ""} {
                lappend collected $pending
                set pending ""
            } else {
                regexp {^(\S+)} $trimmed pname
                lappend collected $pname
            }
            continue
        }

        # name-only continuation line (long path wrapped to its own line)
        set pending $trimmed
    }

    return $collected
}
```

버퍼 삽입&weight 추출

```
proc insert_buffer_with_size_only {pin_list buf_cell} {
    set new_buf [insert_buffer $pin_list $buf_cell]

    if {[sizeof_collection $new_buf] > 0} {
        set_size_only $new_buf
        foreach_in_collection c $new_buf {
            echo "     set_size_only applied to: [get_object_name $c]"
        }
    } else {
        echo "     WARNING: insert_buffer returned no cell collection;"
        echo "     set_size_only was not applied automatically."
        echo "     You may need to locate and mark it manually, e.g.:"
        echo "       set_size_only [get_cells -hier *eco_cell*]"
    }

    return $new_buf
}

proc get_pin_weight {p} {
    # Prefer fanout_load (the metric max_fanout actually checks).
    # Some libraries (e.g. certain TSMC180 kits) do not define
    # fanout_load; DC returns an empty string with just a Warning
    # in that case (not a Tcl error), so check for "" explicitly
    # instead of relying on catch.
    set w [get_attribute -quiet $p fanout_load]
    if {$w == ""} {
        set w [get_attribute -quiet $p pin_capacitance]
    }
    if {$w == ""} {
        set w [get_attribute -quiet $p capacitance]
    }
    if {$w == ""} {
        set w 1.0
    }
    return $w
}
```

load pin 수집

```
proc split_fanout {net_name max_fanout {buf_cell "saed32lvt_tt0p78v125c/NBUFFX2_LVT"}} {

    # ---- 0. check net exists ----
    set net_col [get_nets $net_name]
    if {[sizeof_collection $net_col] == 0} {
        echo "ERROR: net '$net_name' not found."
        return
    }

    # ---- 1. collect load pins and their weight ----
    set load_pins [get_pins -of_objects $net_col -filter "pin_direction==in"]
    set n_loads [sizeof_collection $load_pins]

    if {$n_loads == 0} {
        echo "ERROR: no load pins connected to '$net_name'."
        return
    }

    if {$n_loads <= 1} {
        # The sole load found via get_pins -of_objects may be a
        # hierarchical boundary pin/port. get_nets/all_fanout have
        # both proven unreliable for tracing the real fanout tree
        # through hierarchy in this environment, so fall back to
        # parsing report_net's own text output directly - it has
        # consistently shown the correct real load list.
        set sole_pin [index_collection $load_pins 0]
        set sole_pin_name [get_object_name $sole_pin]

        set parsed_loads [parse_report_net_loads $net_name]
        set n_parsed [llength $parsed_loads]

        if {$n_parsed > 1} {
            echo "================================================="
            echo " '$net_name' has only 1 load pin via get_pins ($sole_pin_name),"
            echo " but report_net -connections shows $n_parsed real load(s)."
            echo " Using report_net's load list directly (count-based split)."
            echo "================================================="
```

역할 — report_net -connections 결과에서 대상 net에 연결된 load pin 목록을 추출해 buffer 분산 계산의 입력으로 사용.

split_fanout (2/3)

insert_buffer 실행

```
    echo " -> divisor = $divisor (max group weight after split: $max_group_sum)"
    echo "================================================="

    # ---- 4. insert_buffer for each group ----
    for {set g 0} {$g < $divisor} {incr g} {
        set pin_list $groups($g)
        if {[llength $pin_list] == 0} { continue }

        set pcount [llength $pin_list]
        echo "---- Group $g : $pcount pins, weight sum = $gsum($g) ----"
        foreach p $pin_list { echo "     $p" }

        insert_buffer_with_size_only $pin_list $buf_cell
    }

    echo "================================================="
    echo " insert_buffer done. Check results with:"
    echo "   report_net -connections $net_name"
    echo "   report_constraint -all_violators -max_fanout"
    echo "   report_constraint -all_violators -max_transition"
    echo "================================================="
}
```

삽입 결과 확인

```
set n [llength $violators]
echo "================================================="
echo " max_fanout violating nets found : $n"
echo " max_fanout value used (input)   : $max_fanout"
echo "================================================="

if {$n == 0} {
    echo "No violating nets found."
    return
}

set idx 1
foreach net_name $violators {
    echo "-------------------------------------------------"
    set label "$idx of $n"
    echo " Processing ($label) : $net_name  (max_fanout=$max_fanout)"
    echo "-------------------------------------------------"

    split_fanout $net_name $max_fanout $buf_cell

    incr idx
}

echo "================================================="
echo " insert_buffer done for all $n net(s)."
echo " Check results with:"
echo "   report_constraint -all_violators -max_fanout"
echo "   report_constraint -all_violators -max_transition"
echo "================================================="
```

전체 violation split_fanout

```
proc fix_all_max_fanout_violations {max_fanout {buf_cell "saed32lvt_tt0p78v125c/NBUFFX2_LVT"}} {

    redirect -variable rpt_text {report_constraint -all_violators -max_fanout}

    set raw_lines [split $rpt_text "\n"]

    # ---- Pre-pass: merge wrapped lines ----
    # When a net path is too long, DC prints it alone on one line and
    # pushes the numeric fanout columns to the next line. Detect a
    # "name-only" line (a single token, no digits, not a header/Total
    # line) and merge it with whatever follows.
    set lines {}
    set pending ""
    foreach raw $raw_lines {
        set trimmed [string trim $raw]
        if {$trimmed == ""} { continue }

        set is_name_only 0
        if {[regexp {^[A-Za-z_][A-Za-z0-9_/\[\].]*$} $trimmed]} {
            if {![string match "*VIOLATED*" $trimmed] && \
                ![string match "*Fanout*" $trimmed]   && \
                ![string match "*Required*" $trimmed]  && \
                ![string match "Total*" $trimmed]      && \
                ![string match "----*" $trimmed]} {
                set is_name_only 1
            }
        }

        if {$is_name_only} {
            set pending $trimmed
            continue
        }

        if {$pending != ""} {
            lappend lines "$pending $trimmed"
            set pending ""
        } else {
            lappend lines $trimmed
        }
    }
```

역할 — 한계를 넘는 net의 load를 나눠 buffer를 삽입. 호출 형태는 split_fanout <net> <max_fanout> <buffer cell>.
fix_all_max_fanout_violations <max_fanout> <buffer cell>으로 모든 violation net에 대해 split_fanout 수행

split_fanout (3/3)

- fix_all_max_fanout_violations_loop

```
proc fix_all_max_fanout_violations_loop {max_fanout {buf_cell "saed32lvt_tt0p78v125c/NBUFFX2_LVT"} {max_iter 10}} {
    for {set i 1} {$i <= $max_iter} {incr i} {
        redirect -variable rpt_check {report_constraint -all_violators -max_fanout}
        if {![string match "*VIOLATED*" $rpt_check]} {
            set iter_done [expr {$i - 1}]
            echo "No more max_fanout violations. Done after $iter_done iteration(s)."
            return
        }
        echo "=== Iteration $i ==="
        fix_all_max_fanout_violations $max_fanout $buf_cell
    }
    echo "Reached max_iter=$max_iter. Some violations may still remain."
    echo "Check with: report_constraint -all_violators -max_fanout"
}
```

역할 — 한계를 넘는 net의 load를 나눠 buffer를 삽입하고, report_constraint에 위반이 없어질 때까지 반복. 호출 형태는 fix_all_max_fanout_violations_loop <max_fanout> <buffer cell>.

report_transition_drivers (1/4)

merge_wrapped_report_lines

```
proc merge_wrapped_report_lines {raw_text} {
    set raw_lines [split $raw_text "\n"]
    set lines {}
    set pending ""

    foreach raw $raw_lines {
        set trimmed [string trim $raw]
        if {$trimmed == ""} { continue }

        set is_name_only 0
        if {[regexp {^[A-Za-z_][A-Za-z0-9_/\[\].]*$} $trimmed]} {
            if {![string match "*VIOLATED*" $trimmed] &&
                ![string match "*Fanout*" $trimmed]    &&
                ![string match "*Transition*" $trimmed] &&
                ![string match "*Required*" $trimmed]   &&
                ![string match "Total*" $trimmed]       &&
                ![string match "----*" $trimmed]} {
                set is_name_only 1
            }
        }

        if {$is_name_only} {
            set pending $trimmed
            continue
        }

        if {$pending != ""} {
            lappend lines "$pending $trimmed"
            set pending ""
        } else {
            lappend lines $trimmed
        }
    }
    return $lines
}

# ---- 1. get violating net names from max_transition report ----
proc get_max_transition_violator_nets {} {
    redirect -variable rpt_text {report_constraint -all_violators -max_transition}

    set lines [merge_wrapped_report_lines $rpt_text]

    set violators {}
    foreach line $lines {
        if {[regexp {^\s*(\S+)\s+[0-9.]+\s+[0-9.]+\s+-?[0-9.]+\s+\(VIOLATED\)} $line \
                match tok]} {
            if {$tok == "PIN"} { continue }
            lappend violators $tok
        }
    }
    return $violators
}
```

report 파싱

```
proc get_net_driver_info {net_name} {
    redirect -variable rpt_net {report_net -connections -verbose $net_name}

    set lines [merge_wrapped_report_lines $rpt_net]

    set driver_pin ""
    set driver_cell ""

    set in_driver 0
    foreach line $lines {
        if {[string match "Driver Pins*" $line]} {
            set in_driver 1
            continue
        }
        if {[string match "Load Pins*" $line]} {
            break
        }
        if {!$in_driver} { continue }
        if {[string match "----*" $line]} { continue }

        # merged line looks like:
        #   DU/DP_OP_16_127_9689/U370/Y Output Pin (NAND2X0_HVT) 0
        if {[regexp {^(\S+)\s+(?:Output|Input)\s+Pin\s+\(([^)]+)\)} $line \
                match pname ctype]} {
            set driver_pin  $pname
            set driver_cell $ctype
            break
        }
    }

    return [list $driver_pin $driver_cell]
}
```

violator net 수집

```
proc report_transition_driver_cells {} {
    set nets [get_max_transition_violator_nets]
    set n [llength $nets]

    echo "================================================="
    echo " max_transition violating nets found : $n"
    echo "================================================="

    if {$n == 0} {
        echo "No violating nets found."
        return
    }

    echo [format "%-45s %-30s %-20s" "Net" "Driver Pin" "Driver Cell"]
    echo [string repeat "-" 95]

    foreach net_name $nets {
        set info [get_net_driver_info $net_name]
        set driver_pin  [lindex $info 0]
        set driver_cell [lindex $info 1]

        if {$driver_pin == ""} {
            set driver_pin  "(not found)"
            set driver_cell "(not found)"
        }

        echo [format "%-45s %-30s %-20s" $net_name $driver_pin $driver_cell]
    }

    echo [string repeat "-" 95]
    echo " Total: $n net(s)"
    echo "================================================="
}
```

역할 — 줄바꿈으로 잘린 report 라인을 합친 뒤 max_transition 위반 net을 모으고 각 net을 구동하는 driver cell을 식별.

report_transition_drivers (2/4)

driver cell 식별

```
proc cell_drive_strength {cell_name} {
    if {![regexp {^.*[XD]([0-9]+)(_.*)?$} $cell_name match digits]} {
        return -1
    }
    return $digits
}

proc upsize_transition_driver_cells {} {
    set nets [get_max_transition_violator_nets]
    set n [llength $nets]

    echo "================================================="
    echo " max_transition violating nets found : $n"
    echo "================================================="

    if {$n == 0} {
        echo "No violating nets found."
        return
    }

    set idx 1
    foreach net_name $nets {
        echo "-------------------------------------------------"
        set label "$idx of $n"
        echo " ($label) Net: $net_name"
```

```
        set info [get_net_driver_info $net_name]
        set driver_pin  [lindex $info 0]
        set driver_cell [lindex $info 1]

        if {$driver_pin == "" || $driver_cell == ""} {
            echo "   SKIP: could not determine driver pin/cell."
            incr idx
            continue
        }

        # strip trailing "/PinName" to get the instance path
        set slash_idx [string last "/" $driver_pin]
        if {$slash_idx < 0} {
            echo "   SKIP: could not parse instance from '$driver_pin'."
            incr idx
            continue
        }
        set inst_path [string range $driver_pin 0 [expr {$slash_idx - 1}]]

        echo "   Driver instance : $inst_path"
        echo "   Current cell    : $driver_cell"

        set pattern [cell_family_pattern $driver_cell]
```

역할 — get_max_transition_violator_nets로 위반 net을 모으고, net마다 get_net_driver_info로 driver pin과 cell을 확인. driver pin 경로에서 instance path를 분리하고, cell_drive_strength로 셀 이름에 붙은 구동 강도 숫자를 추출.

report_transition_drivers (3/4)

- 대체 셀 결정

```
if {$pattern == ""} {
    echo "   SKIP: could not parse drive-strength pattern from '$driver_cell'."
    incr idx
    continue
}

set family [get_lib_cells -quiet "*/${pattern}"]
set n_fam [sizeof_collection $family]
if {$n_fam == 0} {
    echo "   SKIP: no cells found matching pattern '*/${pattern}'."
    incr idx
    continue
}

# build sorted (strength, full_lib_cell_name) list
set fam_list {}
foreach_in_collection c $family {
    set full_name [get_object_name $c]
    set base_name [lindex [split $full_name "/"] end]
    set strength [cell_drive_strength $base_name]
    if {$strength < 0} { continue }
    lappend fam_list [list $strength $full_name $base_name]
}
set sorted_fam [lsort -integer -index 0 $fam_list]
```

```
# find current cell's position in the sorted family
set cur_index -1
for {set i 0} {$i < [llength $sorted_fam]} {incr i} {
    set entry [lindex $sorted_fam $i]
    if {[lindex $entry 2] == $driver_cell} {
        set cur_index $i
        break
    }
}

if {$cur_index < 0} {
    echo "   SKIP: current cell '$driver_cell' not found in its own family list."
    incr idx
    continue
}

if {$cur_index + 1 >= [llength $sorted_fam]} {
    echo "   SKIP: '$driver_cell' is already the largest size in this family."
    incr idx
    continue
}

set next_entry [lindex $sorted_fam [expr {$cur_index + 1}]]
set next_full_name [lindex $next_entry 1]
set next_base_name [lindex $next_entry 2]
```

역할 — get_lib_cells로 같은 family의 셀을 수집해 (구동 강도, 셀 이름) 목록을 만들고 강도 순으로 정렬. 현재 driver의 위치를 찾아 바로 다음 단계 셀을 선택하며, 계열을 찾지 못하거나 이미 최대 크기면 건너뜀.

report_transition_drivers (4/4)

- size_cell 실행

```
        echo "   Upsizing to      : $next_base_name"
        size_cell $inst_path $next_full_name

        incr idx
    }

    echo "=================================================="
    echo " upsize pass done. Check results with:"
    echo "   report_constraint -all_violators -max_transition"
    echo "   report_constraint -all_violators -hold"
    echo "=================================================="
}
```

역할 — size_cell로 driver instance를 한 단계 큰 셀로 교체. 모든 net 처리 후 report_constraint -all_violators로 max_transition과 hold를 다시 확인하도록 안내.

03

SECTION

Synthesis @ 2.0 ns

SAED32 기준 합성 — compile0 이후 발생한 위반을 수동으로 해결.

정태웅 · 청주대학교

Common Setup & Constraints

Common_setup(target/link)

```
set hvt_libs    "saed32hvt_tt0p78v125c.db"

set rvt_libs    "saed32rvt_tt0p78v125c.db"

set lvt_libs    "saed32lvt_tt0p78v125c.db"


set TARGET_LIBRARY_FILES          "$lvt_libs $rvt_libs $hvt_libs"

set MIN_LIBRARY_FILES             "";

set MW_REFERENCE_CONTROL_FILE     "";

set link_library "saed32sram_tt1p05v125c.db"
```

constraints

```
current_design Risc_16_bit

create_clock -period 2.0 -name CLK [get_ports clk]
set_clock_uncertainty -setup 0.080 [get_clocks CLK]
set_clock_uncertainty -hold  0.040 [get_clocks CLK]
set_clock_transition 0.050 [get_clocks CLK]

set_input_transition 0.050 [get_ports rst_n]
set_ideal_network [get_ports rst_n]
#set_dont_touch_network [get_ports rst_n]

create_generated_clock -divide_by 1 -name SRAM_CLK -invert -source [get_ports clk] [get_pins {DU/dm/u_sr
am_high/CE DU/dm/u_sram_low/CE}]
set_clock_uncertainty -setup 0.080 [get_clocks SRAM_CLK]
set_clock_uncertainty -hold  0.040 [get_clocks SRAM_CLK]
set_clock_transition 0.050 [get_clocks SRAM_CLK]


set_output_delay 0.2 -clock CLK [get_ports {done pass}]
set_load 0.01 [get_ports {done pass}]

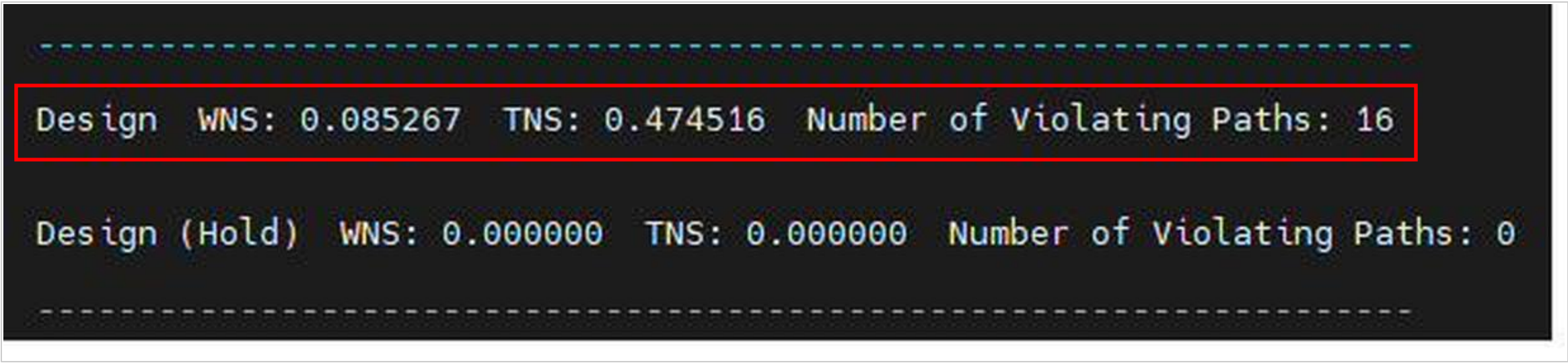
set_multi_vth_constraint -lvth_groups {saed32cell_lvt} -lvth_percentage 27 -type hard

set_max_transition 0.100 [current_design]
set_max_fanout 8 [current_design]
```

compile0 violation ① — Setup Timing

초기 compile 직후 setup 경로에 negative slack 발생.

■ report_qor · compile0 VIOLATED



■ 수치

Setup WNS +0.085 ns · TNS 0.475 · 위반 16 paths (hold 위반 0).

■ 원인

기본 매핑만으로는 일부 경로가 2.0 ns 안에 setup을 만족하지 못함.

■ 대응

timing effort 상향 + mapped path group으로 critical 경로 집중 재최적화.

compile0 violation ② — max_fanout · max_transition

timing violation과 별개로, design rule에서도 max_fanout · max_transition 초과 발생

- report_constraint · DRC VIOLATED

```
max_transition

                          Required        Actual
Net                      Transition     Transition        Slack
-------------------------------------------------------------------
DU/alu_unit/n559         0.100000       0.103358       -0.003358 (VIOLATED)
    PIN :   DU/alu_unit/U395/A1
                         0.100000       0.103358       -0.003358 (VIOLATED)
    PIN :   DU/alu_unit/U195/A1
                         0.100000       0.103358       -0.003358 (VIOLATED)
    PIN :   DU/alu_unit/U336/A1
                         0.100000       0.103358       -0.003358 (VIOLATED)
    PIN :   DU/alu_unit/U180/A3
                         0.100000       0.103358       -0.003358 (VIOLATED)
    PIN :   DU/alu_unit/U445/A1
                         0.100000       0.103358       -0.003358 (VIOLATED)
    PIN :   DU/alu_unit/U169/Y
                         0.100000       0.103358       -0.003358 (VIOLATED)
DU/n162                  0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U42/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U52/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U78/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U122/A1   0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U155/A1   0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U97/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U86/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U81/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U71/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U68/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U63/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U39/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U112/A1   0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U107/A1   0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U102/A1   0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U55/A1    0.100000       0.101137       -0.001137 (VIOLATED)
    PIN :   DU/U25/Y     0.100000       0.101137       -0.001137 (VIOLATED)
-------------------------------------------------------------------
Total                    2                             -0.004495

max_fanout

                          Required        Actual
Net                       Fanout          Fanout          Slack
-------------------------------------------------------------------
DU/alu_unit/DP_OP_19_128_5797/n446
                         8.000000       9.177000       -1.177000 (VIOLATED)
-------------------------------------------------------------------
Total                    1                             -1.177000
```

max_fanout

DP_OP_19_128_5797/n446 fanout ≈ 9 > 8 — 대형 연산기 출력에 부하 집중.

max_transition

alu_unit 다수 pin에서 0.101 – 0.103 > 0.100 초과.

대응

fanout은 buffer 삽입으로 분산, transition은 driver 업사이징.

전체 TCL (1/2)

- dc.tcl (1/3)

```
####################design_read####################
set hdlin_enable_hier_map true
set_svf output/risc.svf
read_verilog ../../rtl/risc/Risc_16bit.v
read_verilog ../../rtl/risc/Datapath.v
read_verilog ../../rtl/risc/Control_Unit.v
read_verilog ../../rtl/risc/ALU.v
read_verilog ../../rtl/risc/alu_control.v
read_verilog ../../rtl/risc/GPRs.v
read_verilog ../../rtl/risc/Instruction_Memory.v
read_verilog ../../rtl/risc/Data_Memory.v

####################design_check####################
current_design Risc_16_bit
link
set_verification_top
check_design
check_library
source risc_con.con
check_timing

####################env_option####################
set_dont_use [get_lib_cells */FADD*]
set_app_var compile_ultra_ungroup_dw false
set_cost_priority -delay
set_auto_disable_drc_nets -on_clock_network true
```

```
####################compile0####################
create_auto_path_groups -mode rtl
compile_ultra -no_autoungroup -no_boundary_optimization
remove_auto_path_groups
report_qor
rt
rc

####################report_before##############
report_area > reports/area.rpt
report_power > reports/power.rpt
report_qor > reports/qor.rpt
report_reference > reports/reference.rpt
rt > reports/timing.rpt
rc > reports/constraints.rpt

####################compile1##################
set_app_var compile_timing_high_effort true
create_auto_path_groups -mode mapped -slack 0.1
compile_ultra -no_autoungroup -no_boundary_optimization -incremental
remove_auto_path_groups
report_qor
rt
rc
```

전체 TCL (2/2)

dc.tcl (2/3)

```
####################compile2##############
set_cost_priority -design_rules
compile_ultra -no_autoungroup -no_boundary_optimization -incremental -only_design_rule
remove_auto_path_groups
report_qor
rt
rc

####################fix_fanout###############
report_net -connection -verbose DU/alu_unit/DP_OP_19_128_5797/n446 > connection_n446.rpt
insert_buffer {     DU/alu_unit/DP_OP_19_128_5797/U178/A2     DU/alu_unit/DP_OP_19_128_5797/U179/A1
DU/alu_unit/DP_OP_19_128_5797/U192/A1     DU/alu_unit/DP_OP_19_128_5797/U223/A1     DU/alu_unit/DP_OP_19
_128_5797/U186/A1     DU/alu_unit/DP_OP_19_128_5797/U193/A2 } saed32lvt_tt0p78v125c/NBUFFX2_LVT
insert_buffer {     DU/alu_unit/DP_OP_19_128_5797/U237/A2     DU/alu_unit/DP_OP_19_128_5797/U230/A1
DU/alu_unit/DP_OP_19_128_5797/U227/A1     DU/alu_unit/DP_OP_19_128_5797/U181/A1     DU/alu_unit/DP_OP_19
_128_5797/U191/A2 } saed32lvt_tt0p78v125c/NBUFFX2_LVT
set_dont_touch [get_cells DU/alu_unit/DP_OP_19_128_5797/eco_cell]
set_dont_touch [get_cells DU/alu_unit/DP_OP_19_128_5797/eco_cell_0]
report_qor
rt
rc
```

dc.tcl (3/3)

```
####################compile3############
create_auto_path_groups -mode mapped -slack 0.05
compile_ultra -no_autoungroup -no_boundary_optimization -incremental
remove_auto_path_groups
report_qor
rt
rc

#######################fix_transition##########
report_net -connections -verbose [get_nets DU/alu_unit/n252]
size_cell DU/alu_unit/U342 saed32hvt_tt0p78v125c/INVX2_HVT
report_net -connections -verbose [get_nets n2]
size_cell U2 saed32rvt_tt0p78v125c/NBUFFX4_RVT
report_net -connections -verbose [get_nets DU/n162]
size_cell DU/U25 saed32rvt_tt0p78v125c/INVX4_RVT
report_net -connections -verbose [get_nets DU/n128]
size_cell DU/U165 saed32hvt_tt0p78v125c/OR4X2_HVT
report_qor
rt
rc

########################report_after##################
report_area > reports/area_fixed.rpt
report_power > reports/power_fixed.rpt
report_qor > reports/qor_fixed.rpt
report_reference > reports/reference_fixed.rpt
rt > reports/timing_fixed.rpt
rc > reports/constraints_fixed.rpt
source ../../tools_cisc/dc/Tcl/vth.tcl > reports/LVT_per.rpt
write_file -format verilog -hierarchy -output output/risc_netlist.v
write_file -format ddc -hierarchy -output output/risc_netlist.ddc
write_sdc output/risc.sdc
write_sdf output/risc.sdf
set_svf -off
history -h > reports/risc_his.rpt
```

결과 상세 — report_qor

report_qor (1)

```
****************************************
Report : qor
Design : Risc_16_bit
Version: S-2021.06-SP5-5
Date   : Wed Jul 29 13:59:37 2026
****************************************


  Timing Path Group 'CLK'
  -----------------------------------
  Levels of Logic:          37.000000
  Critical Path Length:      1.890364
  Critical Path Slack:       0.000306
  Critical Path Clk Period:  2.000000
  Total Negative Slack:      0.000000
  No. of Violating Paths:    0.000000
  Worst Hold Violation:      0.000000
  Total Hold Violation:      0.000000
  No. of Hold Violations:    0.000000
  -----------------------------------

  Timing Path Group 'SRAM_CLK'
  -----------------------------------
  Levels of Logic:          21.000000
  Critical Path Length:      0.959299
  Critical Path Slack:       0.000340
  Critical Path Clk Period:  2.000000
  Total Negative Slack:      0.000000
  No. of Violating Paths:    0.000000
  Worst Hold Violation:      0.000000
  Total Hold Violation:      0.000000
  No. of Hold Violations:    0.000000
  -----------------------------------
```

report_qor (2)

```
Cell Count
----------------------------------
Hierarchical Cell Count:          9
Hierarchical Port Count:        344
Leaf Cell Count:               1569
Buf/Inv Cell Count:             435
Buf Cell Count:                  65
Inv Cell Count:                 370
CT Buf/Inv Cell Count:            1
Combinational Cell Count:      1486
Sequential Cell Count:           83
Macro Count:                      0
----------------------------------


Area
----------------------------------
Combinational Area:     3339.706340
Noncombinational Area:   581.989769
Buf/Inv Area:            662.045125
Total Buffer Area:       150.961540
Total Inverter Area:     511.083585
Macro/Black Box Area:   9003.128906
Net Area:                  0.000000
Net XLength        :   11399.368164
Net YLength        :   10862.843750
----------------------------------
Cell Area:              12924.825015
Design Area:            12924.825015
Net Length         :    22262.210938

Design Rules
----------------------------------
Total Number of Nets:          1634
Nets With Violations:             0
Max Trans Violations:             0
Max Cap Violations:               0
----------------------------------
```

report_qor (3)

```
Hostname: localhost.localdomain

Compile CPU Statistics
-----------------------------------------
Resource Sharing:                 0.000000
Logic Optimization:               0.000000
Mapping Optimization:             9.660866
-----------------------------------------
Overall Compile Time:             22.216690
Overall Compile Wall Clock Time: 23.762432

--------------------------------------------------------------------------------

Design  WNS: 0.000000  TNS: 0.000000  Number of Violating Paths: 0


Design (Hold)  WNS: 0.000000  TNS: 0.000000  Number of Violating Paths: 0

--------------------------------------------------------------------------------
```

결과 상세 — report_timing (1/2)

report_timing (1)

```
****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : Risc_16_bit
Version: S-2021.06-SP5-5
Date   : Wed Jul 29 13:59:37 2026
****************************************

 * Some/all delay information is back-annotated.

Operating Conditions: tt0p78v125c   Library: saed32lvt_tt0p78v125c
Wire Load Model Mode: Inactive.

  Startpoint: DU/pc_current_reg[3]
              (rising edge-triggered flip-flop clocked by CLK)
  Endpoint: DU/pass_r_reg
            (rising edge-triggered flip-flop clocked by CLK)
  Path Group: CLK
  Path Type: max

  Point                                                   Incr       Path
  ---------------------------------------------------------------------------
  clock CLK (rise edge)                                 0.000000   0.000000
  clock network delay (ideal)                           0.000000   0.000000
  DU/pc_current_reg[3]/CLK (DFFARX2_LVT)                0.000000   0.000000 r
  DU/pc_current_reg[3]/QN (DFFARX2_LVT)                 0.095163   0.095163 r
  DU/U13/Y (INVX4_RVT)                                  0.027810 * 0.122973 f
  DU/im/pc[3] (Instruction_Memory)                      0.000000   0.122973 f
  DU/im/U29/Y (INVX2_RVT)                               0.031725 * 0.154698 r
  DU/im/U30/Y (AND4X1_LVT)                              0.053152 * 0.207850 r
  DU/im/U35/Y (OR3X2_LVT)                               0.070407 * 0.278257 r
  DU/im/instruction[14] (Instruction_Memory)            0.000000   0.278257 r
  DU/opcode[2] (Datapath_Unit)                          0.000000   0.278257 r
  control/opcode[2] (Control_Unit)                      0.000000   0.278257 r
  control/U10/Y (INVX2_RVT)                             0.019436 * 0.297693 f
  control/U11/Y (AND2X1_LVT)                            0.035518 * 0.333211 f
  control/U13/Y (AND2X1_LVT)                            0.034768 * 0.367979 f
  control/U14/Y (OR2X2_LVT)                             0.036134 * 0.404113 f
  control/alu_op[0] (Control_Unit)                      0.000000   0.404113 f
  DU/alu_op[0] (Datapath_Unit)                          0.000000   0.404113 f
  DU/ALU_Control_unit/ALUOp[0] (alu_control)            0.000000   0.404113 f
  DU/ALU_Control_unit/U18/Y (INVX1_LVT)                 0.015445 * 0.419559 r
  DU/ALU_Control_unit/U19/Y (AND2X1_LVT)                0.029080 * 0.448638 r
  DU/ALU_Control_unit/U23/Y (AOI21X2_LVT)               0.070208 * 0.518847 f
  DU/ALU_Control_unit/ALU_Cnt[0] (alu_control)          0.000000   0.518847 f
  DU/alu_unit/alu_control[0] (ALU)                      0.000000   0.518847 f
  DU/alu_unit/U546/Y (AND2X2_LVT)                       0.051924 * 0.570771 f
  DU/alu_unit/DP_OP_19_128_5797/I3 (ALU_DP_OP_19_128_5797_1)
                                                        0.000000   0.570771 f
```

report_timing (2)

```
DU/alu_unit/DP_OP_19_128_5797/U346/Y (NBUFFX16_LVT) 0.045577 * 0.616348 f
DU/alu_unit/DP_OP_19_128_5797/U317/Y (XNOR2X2_RVT)  0.102584 * 0.718931 f
DU/alu_unit/DP_OP_19_128_5797/U196/Y (NOR2X0_RVT)   0.097498 * 0.816430 r
DU/alu_unit/DP_OP_19_128_5797/U255/Y (OA21X1_RVT)   0.080741 * 0.897170 r
DU/alu_unit/DP_OP_19_128_5797/U310/Y (INVX1_HVT)    0.052910 * 0.950081 f
DU/alu_unit/DP_OP_19_128_5797/U312/Y (AND2X1_RVT)   0.060444 * 1.010524 f
DU/alu_unit/DP_OP_19_128_5797/U241/Y (OR3X1_LVT)    0.047414 * 1.057938 f
DU/alu_unit/DP_OP_19_128_5797/U329/Y (AO21X1_RVT)   0.052121 * 1.110059 f
DU/alu_unit/DP_OP_19_128_5797/U323/Y (XNOR2X1_LVT)  0.080964 * 1.191023 r
DU/alu_unit/DP_OP_19_128_5797/O1[12] (ALU_DP_OP_19_128_5797_1)
                                                    0.000000   1.191023 r
DU/alu_unit/U364/Y (AO21X1_LVT)                     0.060892 * 1.251915 r
DU/alu_unit/U41/Y (INVX2_RVT)                       0.015207 * 1.267122 f
DU/alu_unit/U31/Y (AND2X1_LVT)                      0.034463 * 1.301585 f
DU/alu_unit/U541/Y (NAND4X0_LVT)                    0.028869 * 1.330455 r
DU/alu_unit/U542/Y (NOR3X2_LVT)                     0.078250 * 1.408705 f
DU/alu_unit/zero (ALU)                              0.000000   1.408705 f
DU/U206/Y (AND2X1_LVT)                              0.043270 * 1.451975 f
DU/DP_OP_16_127_9689/I5 (Datapath_Unit_DP_OP_16_127_9689_1)
                                                    0.000000   1.451975 f
DU/DP_OP_16_127_9689/U311/Y (AND2X1_LVT)            0.045899 * 1.497874 f
DU/DP_OP_16_127_9689/U318/Y (MUX21X2_LVT)           0.052716 * 1.550590 f
DU/DP_OP_16_127_9689/U319/Y (AND2X1_LVT)            0.042006 * 1.592596 f
DU/DP_OP_16_127_9689/U320/Y (INVX2_LVT)             0.031899 * 1.624495 r
DU/DP_OP_16_127_9689/U378/Y (OA22X1_LVT)            0.047897 * 1.672392 r
DU/DP_OP_16_127_9689/U379/Y (XOR2X2_LVT)            0.060934 * 1.733326 f
DU/DP_OP_16_127_9689/O2[15] (Datapath_Unit_DP_OP_16_127_9689_1)
                                                    0.000000   1.733326 f
DU/U160/Y (INVX1_LVT)                               0.017276 * 1.750602 r
DU/U162/Y (AND4X1_LVT)                              0.054251 * 1.804853 r
DU/U175/Y (AND4X1_LVT)                              0.051794 * 1.856647 r
DU/U204/Y (OR2X1_LVT)                               0.033716 * 1.890363 r
DU/pass_r_reg/D (DFFARX1_LVT)                       0.000001 * 1.890364 r
data arrival time                                              1.890364

clock CLK (rise edge)                               2.000000   2.000000
clock network delay (ideal)                         0.000000   2.000000
clock uncertainty                                  -0.080000   1.920000
DU/pass_r_reg/CLK (DFFARX1_LVT)                     0.000000   1.920000 r
library setup time                                 -0.029330   1.890670
data required time                                             1.890670

data required time                                             1.890670
data arrival time                                             -1.890364
---------------------------------------------------------------------------
slack (MET)                                                    0.000306
```

결과 상세 — report_timing (2/2)

report_timing (3)

```
Startpoint: DU/pc_current_reg[3]
            (rising edge-triggered flip-flop clocked by CLK)
Endpoint: DU/dm/u_sram_low
          (rising edge-triggered flip-flop clocked by SRAM_CLK)
Path Group: SRAM_CLK
Path Type: max

Point                                                    Incr       Path
-----------------------------------------------------------------------------
clock CLK (rise edge)                                 0.000000   0.000000
clock network delay (ideal)                           0.000000   0.000000
DU/pc_current_reg[3]/CLK (DFFARX2_LVT)                0.000000   0.000000 r
DU/pc_current_reg[3]/QN (DFFARX2_LVT)                 0.095163   0.095163 r
DU/U13/Y (INVX4_RVT)                                  0.027810 * 0.122973 f
DU/im/pc[3] (Instruction_Memory)                      0.000000   0.122973 f
DU/im/U29/Y (INVX2_RVT)                               0.031725 * 0.154698 r
DU/im/U30/Y (AND4X1_LVT)                              0.053152 * 0.207850 r
DU/im/U35/Y (OR3X2_LVT)                               0.070407 * 0.278257 r
DU/im/instruction[14] (Instruction_Memory)            0.000000   0.278257 r
DU/opcode[2] (Datapath_Unit)                          0.000000   0.278257 r
control/opcode[2] (Control_Unit)                      0.000000   0.278257 r
control/U10/Y (INVX2_RVT)                             0.019436 * 0.297693 f
control/U11/Y (AND2X1_LVT)                            0.035518 * 0.333211 f
control/U13/Y (AND2X1_LVT)                            0.034768 * 0.367979 f
control/U14/Y (OR2X2_LVT)                             0.036134 * 0.404113 f
control/alu_op[0] (Control_Unit)                      0.000000   0.404113 f
DU/alu_op[0] (Datapath_Unit)                          0.000000   0.404113 f
DU/ALU_Control_unit/ALUOp[0] (alu_control)            0.000000   0.404113 f
DU/ALU_Control_unit/U18/Y (INVX1_LVT)                 0.015445 * 0.419559 r
DU/ALU_Control_unit/U19/Y (AND2X1_LVT)                0.029080 * 0.448638 r
DU/ALU_Control_unit/U23/Y (AOI21X2_LVT)               0.070208 * 0.518847 f
DU/ALU_Control_unit/ALU_Cnt[0] (alu_control)          0.000000   0.518847 f
DU/alu_unit/alu_control[0] (ALU)                      0.000000   0.518847 f
DU/alu_unit/U546/Y (AND2X2_LVT)                       0.051924 * 0.570771 f
DU/alu_unit/DP_OP_19_128_5797/I3 (ALU_DP_OP_19_128_5797_1)
                                                      0.000000   0.570771 f
DU/alu_unit/DP_OP_19_128_5797/U221/Y (NBUFFX2_LVT)    0.033792 * 0.604563 f
DU/alu_unit/DP_OP_19_128_5797/eco_cell/Y (NBUFFX2_LVT)
                                                      0.048990 * 0.653553 f
DU/alu_unit/DP_OP_19_128_5797/U193/Y (AND2X1_LVT)     0.054487 * 0.708040 f
DU/alu_unit/DP_OP_19_128_5797/U377/Y (INVX1_LVT)      0.017826 * 0.725866 r
DU/alu_unit/DP_OP_19_128_5797/U378/Y (NAND2X0_LVT)    0.031484 * 0.757350 f
DU/alu_unit/DP_OP_19_128_5797/U380/Y (XNOR2X2_LVT)    0.065461 * 0.822811 r
DU/alu_unit/DP_OP_19_128_5797/O1[0] (ALU_DP_OP_19_128_5797_1)
```

report_timing (4)

```
                                                      0.048990 * 0.653553 f
DU/alu_unit/DP_OP_19_128_5797/U193/Y (AND2X1_LVT)     0.054487 * 0.708040 f
DU/alu_unit/DP_OP_19_128_5797/U377/Y (INVX1_LVT)      0.017826 * 0.725866 r
DU/alu_unit/DP_OP_19_128_5797/U378/Y (NAND2X0_LVT)    0.031484 * 0.757350 f
DU/alu_unit/DP_OP_19_128_5797/U380/Y (XNOR2X2_LVT)    0.065461 * 0.822811 r
DU/alu_unit/DP_OP_19_128_5797/O1[0] (ALU_DP_OP_19_128_5797_1)
                                                      0.000000   0.822811 r
DU/alu_unit/U560/Y (AND2X1_LVT)                       0.031978 * 0.854789 r
DU/alu_unit/U260/Y (OR3X1_LVT)                        0.048587 * 0.903376 r
DU/alu_unit/U326/Y (OR3X2_LVT)                        0.055888 * 0.959264 r
DU/alu_unit/result[0] (ALU)                           0.000000   0.959264 r
DU/dm/mem_access_addr[0] (Data_Memory)                0.000000   0.959264 r
DU/dm/u_sram_low/A[0] (SRAM1RW64x8)                   0.000035 * 0.959299 r
data arrival time                                                0.959299

clock SRAM_CLK (rise edge)                            1.000000   1.000000
clock network delay (ideal)                           0.000000   1.000000
clock uncertainty                                    -0.080000   0.920000
DU/dm/u_sram_low/CE (SRAM1RW64x8)                     0.000000   0.920000 r
library setup time                                    0.039639   0.959639
data required time                                               0.959639
-----------------------------------------------------------------------------
data required time                                               0.959639
data arrival time                                               -0.959299
-----------------------------------------------------------------------------
slack (MET)                                                      0.000340
```

결과 상세 — report_constraint / vth

report_constraint

```
****************************************
Report : constraint
         -all_violators
Design : Risc_16_bit
Version: S-2021.06-SP5-5
Date   : Wed Jul 29 13:59:37 2026
****************************************


   max_leakage_power

                            Required        Actual
   Design                   Leakage Power   Leakage Power     Slack
   ---------------------------------------------------------------------
   Risc_16_bit              0.000000        1252714752.000000
                                                          -1252714752.000000
                                                                    (VIOLATED)
```

vth.tcl · LVT_per

```
LVT_per : 24.3778%, RVT_per : 44.4161%, HVT_per : 31.2061%
```

Formality — RTL ≡ Netlist

합성 netlist가 RTL과 논리적으로 등가임을 확인.

fm_shell · Verification Results SUCCEEDED

```
********************************** Verification Results **************************
********
Verification SUCCEEDED
   ATTENTION: synopsys_auto_setup mode was enabled.
              See Synopsys Auto Setup Summary for details.
---------------------------------------------------------
 Reference design: r:/WORK/Risc_16_bit
 Implementation design: i:/WORK/Risc_16_bit
 119 Passing compare points
--------------------------------------------------------------------------------
--------
Matched Compare Points     BBPin    Loop   BBNet     Cut    Port     DFF     LAT
   TOTAL
--------------------------------------------------------------------------------
--------
Passing (equivalent)          36       0       0       0       2      81       0
     119
Failing (not equivalent)       0       0       0       0       0       0       0
       0
********************************************************************************
```

결과

SUCCEEDED — 119 passing compare points, 0 failing / unmatched.

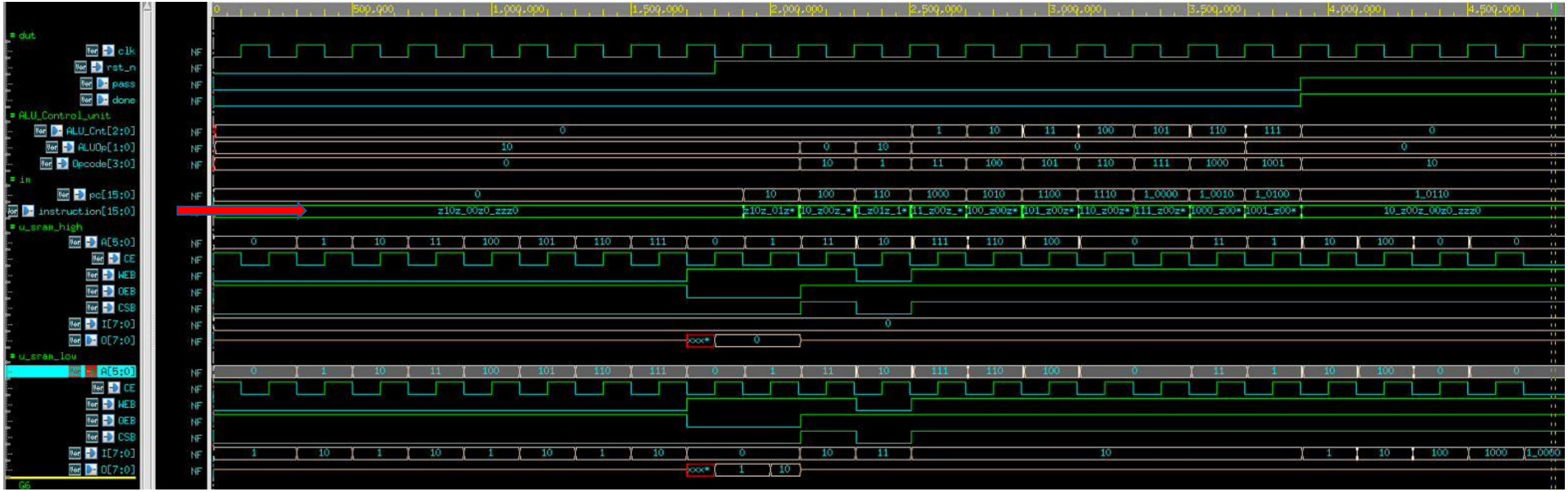
의미

register 최적화 이후에도 RTL 기능이 보존됨 → gate-level 검증으로 진행.

Gate Netlist VCS @ 2.0 ns

gate-level netlist가 RTL과 동일한 done/pass를 재현. But, instruction에서 high-z 발생

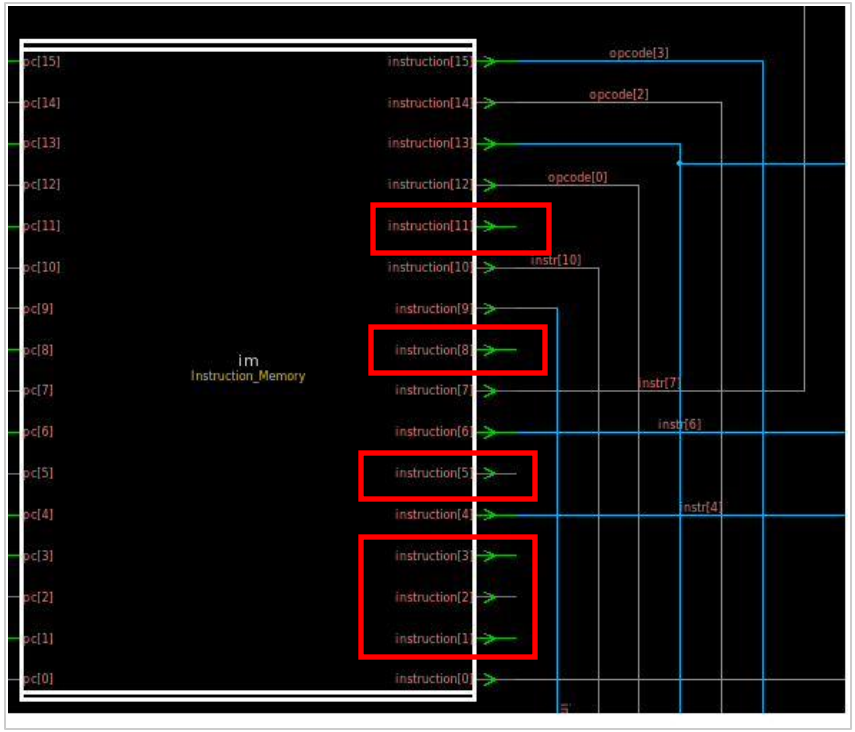
- gate-level waveform (2.0 ns)



instruction 출력의 Z — 이유

미사용 비트라 driver가 없어 gate netlist에서 절단됨.

netlist_vcs · instruction Z



Instruction_Memory · ROM case

```
always @(*) begin
    case (rom_addr)
        4'd0 : instruction_r = 16'h0400; /.
        4'd1 : instruction_r = 16'h0441; /.
        4'd2 : instruction_r = 16'h2050; /.
        4'd3 : instruction_r = 16'h1280; /.
        4'd4 : instruction_r = 16'h3050; /.
        4'd5 : instruction_r = 16'h4050; /.
        4'd6 : instruction_r = 16'h5050; /.
        4'd7 : instruction_r = 16'h6050; /.
        4'd8 : instruction_r = 16'h7050; /.
        4'd9 : instruction_r = 16'h8050; /.
        4'd10: instruction_r = 16'h9050; /.
        4'd11: instruction_r = 16'h2000; /.
        4'd12: instruction_r = 16'hB041; /.
        4'd13: instruction_r = 16'hC040; /.
        4'd14: instruction_r = 16'hD000; /.
        default: instruction_r = 16'h0000;
```

ROM은 사용하는 주소만 case로 정의 → 정의되지 않은 출력 비트에는 driver가 없어 High-Z. 기능에 쓰이지 않는 비트라 done/pass 판정에는 영향 없음.

최종 결과 @ 2.0 ns

setup · hold · DRC 위반 0으로 수렴하고 LVT 목표(< 25 %)까지 충족.

1.890 ns
CLK critical
slack +0.000306

0.959 ns
SRAM_CLK critical
slack +0.000340

12,924.83 μm²
Design area
macro 9,003.13 포함

24.38 %
LVT usage
목표 < 25 % 충족

항목	값
Setup WNS / TNS / 위반 paths	0.000 ns / 0.000 / 0
Hold 위반	0 paths
Max trans / Max cap 위반	0 / 0 (전체 net 1,634 중 위반 0)
Leaf cell / Combinational / Sequential	1,569 / 1,486 / 83
Buf/Inv cell	435 (Buffer 65 · Inverter 370)
Buffer / Inverter area	150.96 / 511.08 μm²
Combinational / Noncombinational area	3,339.71 / 581.99 μm²
Vth 조합	LVT 24.3778 % · RVT 44.4161 % · HVT 31.2061 %
Power (dynamic / leakage / total)	804.49 μW / 1.253 mW / 2.057 mW
Formality	SUCCEEDED · 119 compare points

Reference: Design Compiler · report_qor / report_power / vth.tcl · 정태웅 · 청주대학교

04

SECTION

Challenge @ 1.8 ns

주기를 1.8 ns로 줄여 speed 한계와 그 비용을 확인 — 자동 TCL 적용.

정태웅 · 청주대학교

Constraint 변경 · 2.0 → 1.8 ns

스펙(2.0 ns)은 유지하되 주기만 줄인 spec-up 실험.

risc_con.con (pd1p8) · period 1.8

```
current_design Risc_16_bit

create_clock -period 1.8 -name CLK [get_ports clk]
set_clock_uncertainty -setup 0.080 [get_clocks CLK]
set_clock_uncertainty -hold  0.040 [get_clocks CLK]
set_clock_transition 0.050 [get_clocks CLK]

set_input_transition 0.050 [get_ports rst_n]
set_ideal_network [get_ports rst_n]
#set_dont_touch_network [get_ports rst_n]

create_generated_clock -divide_by 1 -name SRAM_CLK -invert -source [get_ports clk] [get_pins {DU/dm/u_sr
am_high/CE DU/dm/u_sram_low/CE}]
set_clock_uncertainty -setup 0.080 [get_clocks SRAM_CLK]
set_clock_uncertainty -hold  0.040 [get_clocks SRAM_CLK]
set_clock_transition 0.050 [get_clocks SRAM_CLK]


set_output_delay 0.2 -clock CLK [get_ports {done pass}]
set_load 0.01 [get_ports {done pass}]

set_multi_vth_constraint -lvth_groups {saed32cell_lvt} -lvth_percentage 27 -type hard

set_max_transition 0.100 [current_design]
set_max_fanout 8 [current_design]
```

변경점

create_clock -period 1.8 만 수정.

동일 유지

uncertainty · transition · output_delay · design rule · LVT 제약 모두 그대로.

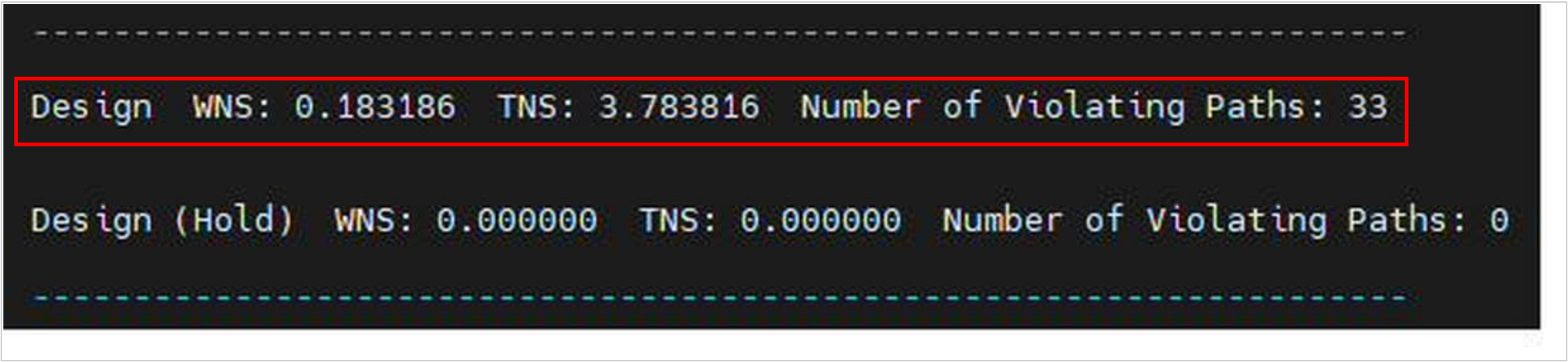
관점

스펙 요구가 아닌 실험 — 같은 RTL · flow 로 얼마나 더 빠르게 닫히는지 확인.

compile0 violation — 위반 급증

주기를 줄이자 setup 위반이 33 paths, WNS 0.183 ns까지 확대.

report_qor · compile0 (1.8 ns) VIOLATED



2.0 ns 대비

위반 16 → 33 paths, WNS 0.085 → 0.183 ns.

worst 경로

DU/CLK_r_REG* 계열 — SRAM_CLK half-cycle 경로가 병목.

대응

자동 TCL과 -retime · driver 업사이징을 반복 적용.

compile0 violation — 상세

setup worst endpoint와 transition 위반 목록.

setup · worst endpoints

```
max_delay/setup ('CLK' group)

                                 Required         Actual
Endpoint                       Path Delay     Path Delay        Slack
-----------------------------------------------------------------------
DU/CLK_r_REG4_S1/D              1.684363       1.867550 r     -0.183186 (VIOLATED)
DU/CLK_r_REG92_S1/D             1.663401       1.837054 r     -0.173653 (VIOLATED)
DU/CLK_r_REG7_S1/D              1.688161       1.861091 f     -0.172930 (VIOLATED)
DU/CLK_r_REG90_S1/D             1.667159       1.833193 r     -0.166034 (VIOLATED)
DU/CLK_r_REG81_S1/D             1.663427       1.829292 r     -0.165865 (VIOLATED)
DU/CLK_r_REG2_S1/D              1.686277       1.844065 f     -0.157788 (VIOLATED)
DU/CLK_r_REG76_S1/D             1.679192       1.832091 f     -0.152899 (VIOLATED)
DU/CLK_r_REG79_S1/D             1.678020       1.830039 f     -0.152020 (VIOLATED)
DU/CLK_r_REG75_S1/D             1.663308       1.814723 r     -0.151415 (VIOLATED)
DU/CLK_r_REG9_S1/D              1.661653       1.811285 r     -0.149632 (VIOLATED)
DU/CLK_r_REG77_S1/D             1.676507       1.824690 f     -0.148183 (VIOLATED)
DU/CLK_r_REG84_S1/D             1.661085       1.808809 r     -0.147724 (VIOLATED)
DU/CLK_r_REG74_S1/D             1.663424       1.808160 r     -0.144736 (VIOLATED)
DU/CLK_r_REG8_S1/D              1.661422       1.805320 r     -0.143898 (VIOLATED)
DU/CLK_r_REG82_S1/D             1.663728       1.806973 r     -0.143245 (VIOLATED)
DU/CLK_r_REG3_S1/D              1.684466       1.825680 r     -0.141214 (VIOLATED)
DU/CLK_r_REG78_S1/D             1.663389       1.798739 r     -0.135349 (VIOLATED)
DU/CLK_r_REG83_S1/D             1.660525       1.794277 r     -0.133752 (VIOLATED)
DU/CLK_r_REG6_S1/D              1.683990       1.814968 f     -0.130977 (VIOLATED)
DU/CLK_r_REG80_S1/D             1.660579       1.790141 r     -0.129562 (VIOLATED)
DU/CLK_r_REG5_S1/D              1.687840       1.810006 r     -0.122166 (VIOLATED)
DU/im/CLK_r_REG91_S1/D          1.683198       1.795678 r     -0.112480 (VIOLATED)
DU/CLK_r_REG86_S1/D             1.643978       1.698593 r     -0.054615 (VIOLATED)
DU/CLK_r_REG85_S1/D             1.640969       1.687552 r     -0.046583 (VIOLATED)
DU/CLK_r_REG87_S1/D             1.643200       1.685050 r     -0.041851 (VIOLATED)
DU/CLK_r_REG88_S1/D             1.644090       1.682541 r     -0.038452 (VIOLATED)
DU/CLK_r_REG0_S1/D              1.661359       1.689896 r     -0.028537 (VIOLATED)


max_delay/setup ('SRAM_CLK' group)

                                 Required         Actual
Endpoint                       Path Delay     Path Delay        Slack
-----------------------------------------------------------------------
DU/dm/u_sram_high/A[2]          0.859999       0.917168 r     -0.057169 (VIOLATED)
DU/dm/u_sram_low/A[2]           0.859999       0.917154 r     -0.057155 (VIOLATED)
DU/dm/u_sram_high/A[0]          0.859793       0.914273 r     -0.054480 (VIOLATED)
DU/dm/u_sram_low/A[0]           0.859793       0.914257 r     -0.054463 (VIOLATED)
DU/dm/u_sram_high/A[1]          0.859561       0.905472 r     -0.045911 (VIOLATED)
DU/dm/u_sram_low/A[1]           0.859561       0.905453 r     -0.045892 (VIOLATED)


max_transition

                                 Required         Actual
Net                            Transition     Transition        Slack
-----------------------------------------------------------------------
DU/reg_file/n113                0.100000       0.166303       -0.066303 (VIOLATED)
    PIN :   DU/reg_file/U158/A1
                                0.100000       0.166303       -0.066303 (VIOLATED)
    PIN :   DU/reg_file/U163/A1
                                0.100000       0.166303       -0.066303 (VIOLATED)
    PIN :   DU/reg_file/U196/A1
```

max_transition 위반

```
                                0.100000       0.101208       -0.001208 (VIOLATED)
DU/n239                         0.100000       0.101012       -0.001012 (VIOLATED)
    PIN :   DU/U5/A             0.100000       0.101012       -0.001012 (VIOLATED)
    PIN :   DU/U238/A4          0.100000       0.101012       -0.001012 (VIOLATED)
    PIN :   DU/CLK_r_REG81_S1/Q
                                0.100000       0.101012       -0.001012 (VIOLATED)
    PIN :   DU/DP_OP_16_127_9689/U305/A
                                0.100000       0.101012       -0.001012 (VIOLATED)
    PIN :   DU/DP_OP_16_127_9689/U331/A2
                                0.100000       0.101012       -0.001012 (VIOLATED)
    PIN :   DU/DP_OP_16_127_9689/U403/A2
                                0.100000       0.101012       -0.001012 (VIOLATED)

-----------------------------------------------------------------------
Total                           20                -0.420769

max_fanout

                                 Required         Actual
Net                               Fanout          Fanout          Slack
-----------------------------------------------------------------------
DU/alu_unit/n504                8.000000       19.250000      -11.250000
                                                                (VIOLATED)

-----------------------------------------------------------------------
Total                           1                 -11.250000

max_capacitance

                                 Required         Actual
Net                           Capacitance     Capacitance       Slack
-----------------------------------------------------------------------
DU/reg_file/n113                16.000000      26.746494      -10.746494
                                                                (VIOLATED)
DU/reg_file/n136                16.000000      24.902763      -8.902763 (VIOLATED)
DU/reg_file/n111                16.000000      20.046535      -4.046535 (VIOLATED)
DU/reg_file/n109                16.000000      18.846554      -2.846554 (VIOLATED)
DU/read_data2[9]                8.000000       8.676426       -0.676426 (VIOLATED)

-----------------------------------------------------------------------
Total                           5                 -27.218773
```

전체 TCL (1/3)

- dc.tcl (1/3)

```
####################design_read####################
set hdlin_enable_hier_map true
set_svf output/risc_pd1p8.svf
read_verilog ../../rtl/risc/Risc_16bit.v
read_verilog ../../rtl/risc/Datapath.v
read_verilog ../../rtl/risc/Control_Unit.v
read_verilog ../../rtl/risc/ALU.v
read_verilog ../../rtl/risc/alu_control.v
read_verilog ../../rtl/risc/GPRs.v
read_verilog ../../rtl/risc/Instruction_Memory.v
read_verilog ../../rtl/risc/Data_Memory.v

####################design_check####################
current_design Risc_16_bit
link
set_verification_top
check_design
source risc_con.con
check_timing

####################env_option####################
set_dont_use [get_lib_cells */FADD*]
set_app_var compile_ultra_ungroup_dw false
set_cost_priority -delay
set_auto_disable_drc_nets -on_clock_network true
set_optimize_registers true
```

```
####################compile0####################
create_auto_path_groups -mode rtl
compile_ultra -no_autoungroup -retime
remove_auto_path_groups
report_qor
rt
rc

#########################report_before####################
report_area > reports/pd1p8_area.rpt
report_power > reports/pd1p8_power.rpt
report_qor > reports/pd1p8_qor.rpt
report_reference > reports/pd1p8_reference.rpt
rt > reports/pd1p8_timing.rpt
rc > reports/pd1p8_constraints.rpt
```

전체 TCL (2/3)

- dc.tcl (2/3)

```
###################fix_fanout&transition1#################
source split_fanout.tcl
source report_transition_drivers.tcl
split_fanout DU/alu_unit/n504 8 saed32lvt_tt0p78v125c/NBUFFX2_LVT
rc
upsize_transition_driver_cells
rc
upsize_transition_driver_cells
rc
upsize_transition_driver_cells
rc
report_qor

###################compile1#################
set_app_var compile_timing_high_effort true
create_auto_path_groups -mode mapped -slack 0.22
compile_ultra -no_autoungroup -retime -incremental
remove_auto_path_groups
report_qor
rt
rc
```

```
###################compile2#################
create_auto_path_groups -mode mapped -slack 0.11
compile_ultra -no_autoungroup -retime -incremental
remove_auto_path_groups
report_qor
rt
rc

###################fix_fanout&transition2#################
fix_all_max_fanout_violations 8 saed32lvt_tt0p78v125c/NBUFFX2_LVT
rc
upsize_transition_driver_cells
rc
upsize_transition_driver_cells
rc
report_qor
```

전체 TCL (3/3)

dc.tcl (3/3)

```
###################compile3#################
create_auto_path_groups -mode mapped -slack 0.041
compile_ultra -no_autoungroup -retime -incremental
remove_auto_path_groups
report_qor
rt
rc

####################fix last transition violation#################
upsize_transition_driver_cells
rc
upsize_transition_driver_cells
rc
report_qor

########################report_after###################
report_area > reports/pd1p8_area_fixed.rpt
report_power > reports/pd1p8_power_fixed.rpt
report_qor > reports/pd1p8_qor_fixed.rpt
report_reference > reports/pd1p8_reference_fixed.rpt
rt > reports/pd1p8_timing_fixed.rpt
rc > reports/pd1p8_constraints_fixed.rpt
source ../../tools_cisc/dc/Tcl/vth.tcl > reports/pd1p8_LVT_per.rpt
write_file -format verilog -hierarchy -output output/pd1p8_risc_netlist.v
write_file -format ddc -hierarchy -output output/pd1p8_risc_netlist.ddc
write_sdc output/pd1p8_risc.sdc
write_sdf output/pd1p8_risc.sdf
set_svf -off
history -h > reports/pd1p8_risc_his.rpt
```

결과 상세 — report_qor

report_qor (1)

```
****************************************
Report : qor
Design : Risc_16_bit
Version: S-2021.06-SP5-5
Date   : Wed Jul 29 15:50:14 2026
****************************************


  Timing Path Group 'CLK'
  -----------------------------------
  Levels of Logic:          33.000000
  Critical Path Length:      1.686160
  Critical Path Slack:       0.000318
  Critical Path Clk Period:  1.800000
  Total Negative Slack:      0.000000
  No. of Violating Paths:    0.000000
  Worst Hold Violation:      0.000000
  Total Hold Violation:      0.000000
  No. of Hold Violations:    0.000000
  -----------------------------------

  Timing Path Group 'SRAM_CLK'
  -----------------------------------
  Levels of Logic:          17.000000
  Critical Path Length:      0.859721
  Critical Path Slack:       0.000098
  Critical Path Clk Period:  1.800000
  Total Negative Slack:      0.000000
  No. of Violating Paths:    0.000000
  Worst Hold Violation:      0.000000
  Total Hold Violation:      0.000000
  No. of Hold Violations:    0.000000
  -----------------------------------


  Cell Count
  -----------------------------------
  Hierarchical Cell Count:          9
  Hierarchical Port Count:        355
  Leaf Cell Count:               1649
  Buf/Inv Cell Count:             451
  Buf Cell Count:                  74
  Inv Cell Count:                 377
  CT Buf/Inv Cell Count:            1
  Combinational Cell Count:      1554
  Sequential Cell Count:           95
  Macro Count:                      0
  -----------------------------------
```

report_qor (2)

```
Area
-----------------------------------
Combinational Area:     3435.264488
Noncombinational Area:   664.840715
Buf/Inv Area:            674.498183
Total Buffer Area:       172.817925
Total Inverter Area:     501.680258
Macro/Black Box Area:   9003.128906
Net Area:                  0.000000
Net XLength        :   11739.613281
Net YLength        :    8040.333984
-----------------------------------
Cell Area:             13103.234110
Design Area:           13103.234110
Net Length         :   19779.947266


Design Rules
-----------------------------------
Total Number of Nets:          1709
Nets With Violations:             0
Max Trans Violations:             0
Max Cap Violations:               0


Hostname: localhost.localdomain

Compile CPU Statistics
---------------------------------------
Resource Sharing:               0.032379
Logic Optimization:             0.000000
Mapping Optimization:           7.369186
---------------------------------------
Overall Compile Time:           19.519783
Overall Compile Wall Clock Time: 21.038660

--------------------------------------------------------------------

Design  WNS: 0.000000  TNS: 0.000000  Number of Violating Paths: 0


Design (Hold)  WNS: 0.000000  TNS: 0.000000  Number of Violating Paths: 0

--------------------------------------------------------------------
```

결과 상세 — report_timing

report_timing (1)

```
****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : Risc_16_bit
Version: S-2021.06-SP5-5
Date   : Wed Jul 29 15:50:14 2026
****************************************

 * Some/all delay information is back-annotated.

Operating Conditions: tt0p78v125c   Library: saed32lvt_tt0p78v125c
Wire Load Model Mode: Inactive.

Startpoint: DU/CLK_r_REG2_S1
            (rising edge-triggered flip-flop clocked by CLK)
Endpoint: DU/CLK_r_REG92_S1
          (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

  Point                                                    Incr       Path
  -----------------------------------------------------------------------------
  clock CLK (rise edge)                                 0.000000   0.000000
  clock network delay (ideal)                           0.000000   0.000000
  DU/CLK_r_REG2_S1/CLK (DFFARX1_LVT)                    0.000000   0.000000 r
  DU/CLK_r_REG2_S1/QN (DFFARX1_LVT)                     0.077412   0.077412 f
  DU/U172/Y (INVX2_LVT)                                 0.027685 * 0.105097 r
  DU/opcode[3] (Datapath_Unit)                          0.000000   0.105097 r
  control/opcode[3] (Control_Unit)                      0.000000   0.105097 r
  control/U9/Y (NOR3X2_LVT)                             0.077286 * 0.182384 f
  control/alu_op[1] (Control_Unit)                      0.000000   0.182384 f
  DU/alu_op[1] (Datapath_Unit)                          0.000000   0.182384 f
  DU/ALU_Control_unit/IN0 (alu_control)                 0.000000   0.182384 f
  DU/ALU_Control_unit/U14/Y (OR2X1_RVT)                 0.065134 * 0.247518 f
  DU/ALU_Control_unit/U17/Y (NOR3X0_LVT)                0.055102 * 0.302620 r
  DU/ALU_Control_unit/U4/Y (OR2X1_LVT)                  0.036345 * 0.338965 r
  DU/ALU_Control_unit/ALU_Cnt[1] (alu_control)          0.000000   0.338965 r
  DU/alu_unit/alu_control[1] (ALU)                      0.000000   0.338965 r
  DU/alu_unit/U204/Y (OR3X2_LVT)                        0.064650 * 0.403615 r
  DU/alu_unit/U121/Y (INVX4_LVT)                        0.010869 * 0.414484 f
  DU/alu_unit/DP_OP_19_128_5797/I3 (ALU_DP_OP_19_128_5797_1)
                                                        0.000000   0.414484 f
  DU/alu_unit/DP_OP_19_128_5797/eco_cell_1_0/Y (NBUFFX2_LVT)
                                                        0.035112 * 0.449595 f
  DU/alu_unit/DP_OP_19_128_5797/eco_cell_2_0/Y (NBUFFX8_LVT)
                                                        0.040757 * 0.490353 f
```

report_timing (2)

```
DU/alu_unit/DP_OP_19_128_5797/U269/Y (XOR2X1_LVT)   0.069149 * 0.559501 r
DU/alu_unit/DP_OP_19_128_5797/U270/Y (INVX1_RVT)    0.019302 * 0.578804 f
DU/alu_unit/DP_OP_19_128_5797/U312/Y (AND2X1_RVT)   0.066349 * 0.645152 f
DU/alu_unit/DP_OP_19_128_5797/U300/Y (OA21X1_RVT)   0.081535 * 0.726687 f
DU/alu_unit/DP_OP_19_128_5797/U179/Y (OA21X1_RVT)   0.073447 * 0.800134 f
DU/alu_unit/DP_OP_19_128_5797/U325/Y (OA21X1_LVT)   0.046279 * 0.846412 f
DU/alu_unit/DP_OP_19_128_5797/U257/Y (INVX1_LVT)    0.016474 * 0.862886 r
DU/alu_unit/DP_OP_19_128_5797/U309/Y (AO21X1_LVT)   0.036684 * 0.899570 r
DU/alu_unit/DP_OP_19_128_5797/U308/Y (XNOR2X2_LVT)  0.061929 * 0.961499 r
DU/alu_unit/DP_OP_19_128_5797/O1[14] (ALU_DP_OP_19_128_5797_1)
                                                    0.000000   0.961499 r
DU/alu_unit/U516/Y (AO21X1_LVT)                     0.049126 * 1.010625 r
DU/alu_unit/U465/Y (OR3X1_LVT)                      0.045927 * 1.056552 r
DU/alu_unit/U502/Y (INVX1_LVT)                      0.010284 * 1.066836 f
DU/alu_unit/U450/Y (NAND4X0_LVT)                    0.031415 * 1.098251 r
DU/alu_unit/U448/Y (INVX1_LVT)                      0.008948 * 1.107199 f
DU/alu_unit/U270/Y (AND4X1_LVT)                     0.074022 * 1.181221 f
DU/alu_unit/zero (ALU)                              0.000000   1.181221 f
DU/U248/Y (AND2X2_LVT)                              0.063047 * 1.244269 f
DU/DP_OP_16_127_9689/I5 (Datapath_Unit_DP_OP_16_127_9689_2)
                                                    0.000000   1.244269 f
DU/DP_OP_16_127_9689/U410/Y (AO22X1_LVT)            0.053382 * 1.297651 f
DU/DP_OP_16_127_9689/U314/Y (OR2X2_RVT)             0.068195 * 1.365846 f
DU/DP_OP_16_127_9689/U343/Y (AND2X1_LVT)            0.050529 * 1.416375 f
DU/DP_OP_16_127_9689/U383/Y (AOI21X1_LVT)           0.069230 * 1.485605 r
DU/DP_OP_16_127_9689/U384/Y (XOR2X1_LVT)            0.072740 * 1.558346 f
DU/DP_OP_16_127_9689/O2[8] (Datapath_Unit_DP_OP_16_127_9689_2)
                                                    0.000000   1.558346 f
DU/U64/Y (OR2X1_RVT)                                0.058982 * 1.617328 f
DU/U80/Y (OR3X1_LVT)                                0.037031 * 1.654359 f
DU/U36/Y (AO22X1_LVT)                               0.031798 * 1.686157 f
DU/CLK_r_REG92_S1/D (DFFASX1_LVT)                   0.000002 * 1.686159 f
data arrival time                                              1.686159

clock CLK (rise edge)                               1.800000   1.800000
clock network delay (ideal)                         0.000000   1.800000
clock uncertainty                                  -0.080000   1.720000
DU/CLK_r_REG92_S1/CLK (DFFASX1_LVT)                 0.000000   1.720000 r
library setup time                                 -0.033522   1.686478
data required time                                             1.686478
-----------------------------------------------------------------------------
data required time                                             1.686478
data arrival time                                             -1.686159
-----------------------------------------------------------------------------
slack (MET)                                                    0.000319
```

report_timing (3)

```
Startpoint: DU/CLK_r_REG7_S1
            (rising edge-triggered flip-flop clocked by CLK)
Endpoint: DU/dm/u_sram_high
          (rising edge-triggered flip-flop clocked by SRAM_CLK)
Path Group: SRAM_CLK
Path Type: max

Point                                                    Incr       Path
-----------------------------------------------------------------------------
clock CLK (rise edge)                                 0.000000   0.000000
clock network delay (ideal)                           0.000000   0.000000
DU/CLK_r_REG7_S1/CLK (DFFARX2_LVT)                    0.000000   0.000000 r
DU/CLK_r_REG7_S1/QN (DFFARX2_LVT)                     0.096798   0.096798 r
DU/U33/Y (INVX4_RVT)                                  0.027351 * 0.124149 f
DU/reg_file/reg_read_addr_2[0] (GPRs)                 0.000000   0.124149 f
DU/reg_file/U4/Y (NOR2X4_LVT)                         0.086665 * 0.210814 r
DU/reg_file/U252/Y (AO22X1_LVT)                       0.042994 * 0.253808 r
DU/reg_file/U254/Y (OR3X2_LVT)                        0.050763 * 0.304571 r
DU/reg_file/reg_read_data_2[1] (GPRs)                 0.000000   0.304571 r
DU/U171/Y (AND2X2_LVT)                                0.048699 * 0.353270 r
DU/alu_unit/b[1] (ALU)                                0.000000   0.353270 r
DU/alu_unit/U262/Y (NBUFFX4_RVT)                      0.068566 * 0.421837 r
DU/alu_unit/U484/Y (AND3X1_RVT)                       0.073277 * 0.495114 r
DU/alu_unit/U289/Y (NAND2X0_LVT)                      0.020396 * 0.515510 f
DU/alu_unit/U455/Y (NAND3X0_LVT)                      0.035332 * 0.550842 r
DU/alu_unit/U426/Y (AO22X1_LVT)                       0.044599 * 0.595441 r
DU/alu_unit/U459/Y (AOI22X1_LVT)                      0.051244 * 0.646685 f
DU/alu_unit/U562/Y (OR2X1_LVT)                        0.030980 * 0.677665 f
DU/alu_unit/U271/Y (AO22X1_LVT)                       0.032132 * 0.709797 f
DU/alu_unit/U486/Y (OA21X1_LVT)                       0.040898 * 0.750696 f
DU/alu_unit/U487/Y (INVX1_LVT)                        0.018953 * 0.769648 r
DU/alu_unit/U558/Y (AND2X1_LVT)                       0.030669 * 0.800317 r
DU/alu_unit/U492/Y (OR3X2_LVT)                        0.059318 * 0.859635 r
DU/alu_unit/result[0] (ALU)                           0.000000   0.859635 r
DU/dm/mem_access_addr[0] (Data_Memory)                0.000000   0.859635 r
DU/dm/u_sram_high/A[0] (SRAM1RW64x8)                  0.000086 * 0.859721 r
data arrival time                                              0.859721

clock SRAM_CLK (rise edge)                            0.900000   0.900000
clock network delay (ideal)                           0.000000   0.900000
clock uncertainty                                    -0.080000   0.820000
DU/dm/u_sram_high/CE (SRAM1RW64x8)                    0.000000   0.820000 r
library setup time                                    0.039819   0.859819
data required time                                             0.859819
-----------------------------------------------------------------------------
data required time                                             0.859819
data arrival time                                             -0.859721
-----------------------------------------------------------------------------
slack (MET)                                                    0.000098
```

결과 상세 — report_constraint / vth

report_constraint

```
****************************************
Report : constraint
         -all_violators
Design : Risc_16_bit
Version: S-2021.06-SP5-5
Date   : Wed Jul 29 15:50:14 2026
****************************************


   max_leakage_power

                            Required         Actual
   Design                   Leakage Power    Leakage Power     Slack
   -----------------------------------------------------------------
   Risc_16_bit              0.000000         1462087552.000000
                                                              -1462087552.000000
                                                                        (VIOLATED)
```

vth.tcl · LVT_per

```
LVT_per : 26.4724%, RVT_per : 50.5161%, HVT_per : 23.0115%
```

LVT 비율이 24.38 % → 26.47 %로 상승해 목표(< 25 %)를 초과 — trade-off

Formality @ 1.8 ns

1.8 ns netlist도 RTL과 논리적으로 등가.

■ fm_shell · Verification Results **SUCCEEDED**

```
************************************ Verification Results ************************************
Verification SUCCEEDED
   ATTENTION: synopsys_auto_setup mode was enabled.
              See Synopsys Auto Setup Summary for details.
---------------------------------------------------------
 Reference design: r:/WORK/Risc_16_bit
 Implementation design: i:/WORK/Risc_16_bit
 216 Passing compare points
---------------------------------------------------------------------------------------------
Matched Compare Points     BBPin    Loop    BBNet     Cut     Port     DFF      LAT    TOTAL
---------------------------------------------------------------------------------------------
Passing (equivalent)         124       0        0       0        2      90        0      216
Failing (not equivalent)       0       0        0       0        0       0        0        0
*********************************************************************************************
```

■ **결과**

SUCCEEDED — 216 passing compare points, 0 failing.

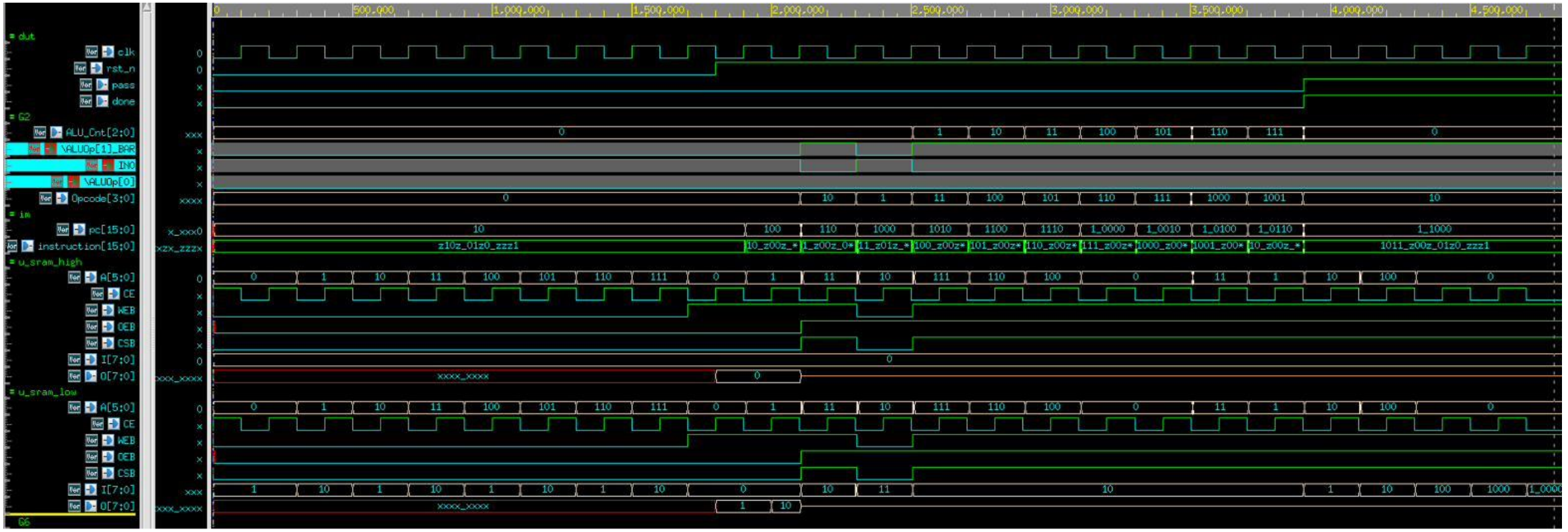
■ **의미**

-retime으로 register가 재배치된 뒤에도 기능 등가 유지.

Gate Netlist VCS @ 1.8 ns

1.8 ns netlist도 gate-level에서 done/pass 재현. But 2.0ns netlist와는 다르게 aluop가 \ALUOp[1]_BAR와 \ALUOp[0], IN0로 나뉨

- gate-level waveform (1.8 ns)



alu_op[1] 신호 매핑 확인

RTL의 alu_op[1]이 netlist에서 IN0 입력과 반전 신호로 구현됨.

netlist · alu_op[1] highlight



관찰

alu_control_unit에서 alu_op[0]은 그대로 보이지만 alu_op[1]은 바로 보이지 않음.

확인

하이라이트된 net이 alu_op[1] — IN0으로 입력되고 인버터를 거쳐 alu_op[1]_BAR로도 연결.

결론

합성 과정의 신호 재구현일 뿐 기능은 동일 — done/pass에 영향 없음.

최종 결과 @ 1.8 ns

1.8 ns에서도 위반 0으로 수렴, 다만 LVT 목표는 초과 — 명확한 트레이드오프.

1.686 ns
CLK critical
slack +0.000318

0.860 ns
SRAM_CLK critical
slack +0.000098

13,103.23 μm²
Design area
2.0 ns 대비 +1.4 %

26.47 %
LVT usage
목표 25 % 초과

항목	값
Setup WNS / TNS / 위반 paths	0.000 ns / 0.000 / 0
Hold 위반	0 paths
Max trans / Max cap 위반	0 / 0
Leaf cell / Combinational / Sequential	1,649 / 1,554 / 95
Buf/Inv cell	451 (Buffer 74 · Inverter 377)
Buffer / Inverter area	172.82 / 501.68 μm²
Combinational / Noncombinational area	3,435.26 / 664.84 μm²
Vth 조합	LVT 26.4724 % · RVT 50.5161 % · HVT 23.0115 %
Power (dynamic / leakage / total)	977.05 μW / 1.462 mW / 2.439 mW
Formality	SUCCEEDED · 216 compare points

Reference: Design Compiler · report_qor / report_power / vth.tcl · 정태웅 · 청주대학교

2.0 ns vs 1.8 ns

동일 RTL · flow · SAED32 corner에서 create_clock -period만 변경. 두 주기 모두 위반 0으로 수렴.

항목	2.0 ns (스펙)	1.8 ns (spec-up)
CLK critical / slack	1.890 ns / +0.000306	1.686 ns / +0.000318
SRAM_CLK critical / slack	0.959 ns / +0.000340	0.860 ns / +0.000098
Setup / Hold 위반 paths	0 / 0	0 / 0
Max trans / Max cap 위반	0 / 0	0 / 0
Leaf cell / Comb / Seq	1,569 / 1,486 / 83	1,649 / 1,554 / 95
Buf · Inv cell	435 (Buf 65 · Inv 370)	451 (Buf 74 · Inv 377)
Design area	12,924.83 μm²	13,103.23 μm² (+1.4 %)
Vth (LVT / RVT / HVT)	24.38 / 44.42 / 31.21 %	26.47 / 50.52 / 23.01 %
Total power	2.057 mW	2.439 mW
Formality	PASS · 119 points	PASS · 216 points

주기를 200 ps 줄여 얻은 속도의 대가로 LVT가 목표(< 25 %)를 넘고 area · 전력이 함께 증가.

pd1p7 (1.7 ns) — 수렴 실패

주기를 1.7 ns로 더 줄이면 setup 위반이 끝까지 남음.

setup 위반 (1) VIOLATED

```
****************************************
Report : constraint
         -all_violators
Design : Risc_16_bit
Version: S-2021.06-SP5-5
Date   : Wed Jul 29 17:07:21 2026
****************************************


  max_delay/setup ('CLK' group)

                             Required        Actual
  Endpoint                   Path Delay      Path Delay        Slack
  -----------------------------------------------------------------
  DU/CLK_r_REG6_S1/D         1.589486        1.752542 r      -0.163056 (VIOLATED)
  DU/CLK_r_REG4_S1/D         1.585755        1.747153 r      -0.161399 (VIOLATED)
  DU/CLK_r_REG110_S1/D       1.565368        1.724167 r      -0.158798 (VIOLATED)
  DU/CLK_r_REG11_S1/D        1.554430        1.713143 r      -0.158713 (VIOLATED)
  DU/CLK_r_REG7_S1/D         1.587835        1.745598 r      -0.157763 (VIOLATED)
  DU/CLK_r_REG5_S1/D         1.554683        1.710683 r      -0.156000 (VIOLATED)
  DU/CLK_r_REG10_S1/D        1.561710        1.714065 r      -0.152355 (VIOLATED)
  DU/CLK_r_REG107_S1/D       1.567233        1.717626 r      -0.150393 (VIOLATED)
  control/CLK_r_REG0_S1/D    1.562444        1.709956 r      -0.147512 (VIOLATED)
  control/CLK_r_REG1_S1/D    1.570298        1.715463 r      -0.145165 (VIOLATED)
  DU/CLK_r_REG102_S1/D       1.554304        1.694243 r      -0.139939 (VIOLATED)
  DU/CLK_r_REG94_S1/D        1.561651        1.700320 r      -0.138669 (VIOLATED)
  DU/CLK_r_REG100_S1/D       1.561065        1.698647 r      -0.137582 (VIOLATED)
  DU/CLK_r_REG99_S1/D        1.561861        1.697151 r      -0.135291 (VIOLATED)
  DU/CLK_r_REG101_S1/D       1.561368        1.695986 r      -0.134618 (VIOLATED)
  DU/CLK_r_REG97_S1/D        1.561535        1.696123 r      -0.134588 (VIOLATED)
  DU/CLK_r_REG9_S1/D         1.587936        1.721255 f      -0.133320 (VIOLATED)
  DU/CLK_r_REG96_S1/D        1.564260        1.690248 r      -0.125987 (VIOLATED)
  DU/CLK_r_REG92_S1/D        1.562299        1.687521 r      -0.125223 (VIOLATED)
  DU/CLK_r_REG93_S1/D        1.564298        1.686659 r      -0.122361 (VIOLATED)
  DU/CLK_r_REG95_S1/D        1.563993        1.682398 r      -0.118406 (VIOLATED)
  DU/CLK_r_REG8_S1/D         1.588004        1.684886 r      -0.096882 (VIOLATED)
  DU/im/CLK_r_REG109_S1/D    1.584070        1.665346 r      -0.081276 (VIOLATED)
  DU/CLK_r_REG98_S1/D        1.561957        1.618971 r      -0.057014 (VIOLATED)
  DU/CLK_r_REG2_S1/D         1.561206        1.610158 r      -0.048953 (VIOLATED)
  DU/CLK_r_REG106_S1/D       1.554099        1.596417 r      -0.042318 (VIOLATED)
  DU/CLK_r_REG105_S1/D       1.544927        1.570981 r      -0.026054 (VIOLATED)
  DU/CLK_r_REG103_S1/D       1.561091        1.576125 r      -0.015034 (VIOLATED)
  DU/reg_file/CLK_r_REG78_S1/D
                             1.570186        1.577746 f      -0.007561 (VIOLATED)
  DU/reg_file/CLK_r_REG79_S1/D
                             1.571670        1.578625 f      -0.006955 (VIOLATED)
  DU/reg_file/CLK_r_REG86_S1/D
                             1.570927        1.577727 f      -0.006799 (VIOLATED)
  DU/reg_file/CLK_r_REG88_S1/D
                             1.571288        1.578047 f      -0.006759 (VIOLATED)
  DU/reg_file/CLK_r_REG89_S1/D
                             1.572061        1.578715 f      -0.006654 (VIOLATED)
```

setup 위반 (2)

```
DU/reg_file/CLK_r_REG87_S1/D
                           1.572087        1.577530 f      -0.005443 (VIOLATED)


  max_delay/setup ('SRAM_CLK' group)

                             Required        Actual
  Endpoint                   Path Delay      Path Delay        Slack
  -----------------------------------------------------------------
  DU/dm/u_sram_low/A[0]      0.809769        0.928483 r      -0.118714 (VIOLATED)
  DU/dm/u_sram_high/A[0]     0.809769        0.928467 r      -0.118698 (VIOLATED)
  DU/dm/u_sram_low/A[1]      0.809503        0.913196 r      -0.103693 (VIOLATED)
  DU/dm/u_sram_high/A[1]     0.809503        0.913189 r      -0.103686 (VIOLATED)
  DU/dm/u_sram_low/A[2]      0.809428        0.912582 r      -0.103154 (VIOLATED)
  DU/dm/u_sram_high/A[2]     0.809428        0.912575 r      -0.103147 (VIOLATED)


  max_transition

                             Required        Actual
  Net                        Transition      Transition        Slack
  -----------------------------------------------------------------
  DU/reg_read_data_1[5]      0.100000        0.110959        -0.010959 (VIOLATED)
      PIN :   DU/alu_unit/DP_OP_19_128_5797/U228/A
                             0.100000        0.110959        -0.010959 (VIOLATED)
      PIN :   DU/alu_unit/DP_OP_19_128_5797/U216/A2
                             0.100000        0.110959        -0.010959 (VIOLATED)
      PIN :   DU/alu_unit/U105/A1
                             0.100000        0.110959        -0.010959 (VIOLATED)
      PIN :   DU/alu_unit/U482/A2
                             0.100000        0.110959        -0.010959 (VIOLATED)
      PIN :   DU/alu_unit/U29/A2
                             0.100000        0.110959        -0.010959 (VIOLATED)
      PIN :   DU/alu_unit/U487/A2
                             0.100000        0.110959        -0.010959 (VIOLATED)
      PIN :   DU/alu_unit/U213/A1
                             0.100000        0.110959        -0.010959 (VIOLATED)
      PIN :   DU/alu_unit/U22/A
                             0.100000        0.110959        -0.010959 (VIOLATED)
      PIN :   DU/reg_file/U115/Y
                             0.100000        0.110959        -0.010959 (VIOLATED)
  DU/reg_file/n104           0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U209/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U171/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U197/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U161/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
```

setup 위반 (3)

```
      PIN :   DU/reg_file/U179/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U186/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U228/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U225/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U249/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U36/A
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U237/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U233/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U192/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U240/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U164/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U204/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U141/A1
                             0.100000        0.104761        -0.004761 (VIOLATED)
      PIN :   DU/reg_file/U137/Y
                             0.100000        0.104761        -0.004761 (VIOLATED)
  -----------------------------------------------------------------
  Total                      2                               -0.015720

  max_leakage_power

                             Required        Actual
  Design                     Leakage Power   Leakage Power     Slack
  -----------------------------------------------------------------
  Risc_16_bit                0.000000        1561429888.000000
                                                          -1561429888.000000
                                                                    (VIOLATED)
```

DU/CLK_r_REG*_S1/D 계열에서 worst −0.163 ns — half-cycle SRAM_CLK 경로가 1.7 ns를 허용하지 않는 구조적 한계.

pd1p7 — report_qor

report_qor (1)

```
****************************************
Report : qor
Design : Risc_16_bit
Version: S-2021.06-SP5-5
Date   : Wed Jul 29 17:07:21 2026
****************************************


  Timing Path Group 'CLK'
  -----------------------------------
  Levels of Logic:          34.000000
  Critical Path Length:      1.752542
  Critical Path Slack:      -0.163056
  Critical Path Clk Period:  1.700000
  Total Negative Slack:     -3.404837
  No. of Violating Paths:   34.000000
  Worst Hold Violation:      0.000000
  Total Hold Violation:      0.000000
  No. of Hold Violations:    0.000000
  -----------------------------------

  Timing Path Group 'SRAM_CLK'
  -----------------------------------
  Levels of Logic:          18.000000
  Critical Path Length:      0.928483
  Critical Path Slack:      -0.118714
  Critical Path Clk Period:  1.700000
  Total Negative Slack:     -0.651092
  No. of Violating Paths:    6.000000
  Worst Hold Violation:      0.000000
  Total Hold Violation:      0.000000
  No. of Hold Violations:    0.000000
  -----------------------------------


  Cell Count
  -----------------------------------
  Hierarchical Cell Count:          9
  Hierarchical Port Count:        362
  Leaf Cell Count:               1724
  Buf/Inv Cell Count:             494
  Buf Cell Count:                  81
  Inv Cell Count:                 413
  CT Buf/Inv Cell Count:            1
  Combinational Cell Count:      1611
  Sequential Cell Count:          113
  Macro Count:                      0
  -----------------------------------
```

report_qor (2)

```
Area
-----------------------------------
Combinational Area:     3606.557546
Noncombinational Area:   793.945869
Buf/Inv Area:            802.840903
Total Buffer Area:       200.773766
Total Inverter Area:     602.067137
Macro/Black Box Area:   9003.128906
Net Area:                 0.000000
Net XLength        :   10591.789062
Net YLength        :   10616.641602
-----------------------------------
Cell Area:             13403.632321
Design Area:           13403.632321
Net Length         :   21208.429688

Design Rules
-----------------------------------
Total Number of Nets:          1791
Nets With Violations:             2
Max Trans Violations:             2
Max Cap Violations:               0
-----------------------------------

Hostname: localhost.localdomain

Compile CPU Statistics
----------------------------------------
Resource Sharing:               0.031281
Logic Optimization:             0.000000
Mapping Optimization:           54.827515
----------------------------------------
Overall Compile Time:           66.621338
Overall Compile Wall Clock Time: 67.898453

--------------------------------------------------------------------

Design  WNS: 0.163056  TNS: 4.055929  Number of Violating Paths: 40


Design (Hold)  WNS: 0.000000  TNS: 0.000000  Number of Violating Paths: 0

--------------------------------------------------------------------
```

1.8 ns vs 1.7 ns — 수렴 / 미수렴 대조

같은 RTL · flow · 자동 TCL을 적용했으나 1.7 ns는 닫히지 않음.

항목	1.8 ns (수렴)	1.7 ns (미수렴)
CLK critical / slack	1.686 ns / +0.000318	1.753 ns / −0.163056
CLK 위반 paths / TNS	0 / 0.000	34 / −3.404837
SRAM_CLK critical / slack	0.860 ns / +0.000098	0.928 ns / −0.118714
SRAM_CLK 위반 paths / TNS	0 / 0.000	6 / −0.651092
Max trans 위반	0	2
Leaf cell / Buf·Inv cell	1,649 / 451	1,724 / 494
Design area	13,103.23 μm²	13,403.63 μm²
Total power	2.439 mW	2.690 mW

1.7 ns에서는 셀과 buffer를 더 투입하고 area · power가 늘었는데도 위반이 남음 — 최적화로 더 밀어낼 수 없는 지점. 따라서 SAED32에서의 주기 하한은 1.8 ns.

05

SECTION

TSMC 180 nm

동일 RTL을 TSMC 180 nm로 이식 — 6.5 ns 수렴 vs 6.4 ns 미수렴.

정태웅 · 청주대학교

Common Setup & constraints

common_setup

```
set TARGET_LIBRARY_FILES          "tcb018g3d3wc.db"  ;#  Targ

#set TARGET_LIBRARY_FILES          "saed32rvt_ss0p95v125c.db"
set MIN_LIBRARY_FILES             "";

set MW_REFERENCE_CONTROL_FILE     "";

set link_library "saed32sram_tt1p05v125c.db"
```

constraints

```
current_design Risc_16_bit

create_clock -period 6.5 -name CLK [get_ports clk]
set_clock_uncertainty -setup 0.080 [get_clocks CLK]
set_clock_uncertainty -hold  0.040 [get_clocks CLK]
set_clock_transition 0.050 [get_clocks CLK]

set_input_transition 0.050 [get_ports rst_n]
set_ideal_network [get_ports rst_n]
#set_dont_touch_network [get_ports rst_n]

create_generated_clock -divide_by 1 -name SRAM_CLK -invert -source [get_ports clk
] [get_pins {DU/dm/u_sram_high/CE DU/dm/u_sram_low/CE}]
set_clock_uncertainty -setup 0.080 [get_clocks SRAM_CLK]
set_clock_uncertainty -hold  0.040 [get_clocks SRAM_CLK]
set_clock_transition 0.050 [get_clocks SRAM_CLK]

set_output_delay 0.2 -clock CLK [get_ports {done pass}]
set_load 0.00001 [get_ports {done pass}]

set_max_transition 3.0340 [current_design]
set_max_fanout 10 [current_design]
```

max_transition — 라이브러리 기준

사용 라이브러리의 default_max_transition을 확인해 그 값으로 설정.

항목	SAED32 32 nm	TSMC 180 nm
Standard cell library	SAED32 (tt0p78v125c)	tcb018g3d3wc
max_transition 기준	0.100 ns (spec 제약)	**lib default_max_transition**
값의 출처	RISC16 spec · set_max_transition 0.100	**라이브러리 header의 default_max_transition 확인**
max_fanout	8	10

이유

32 nm 기준(0.100 ns)은 180 nm 셀 특성에 비해 지나치게 엄격.

적용

라이브러리가 제시하는 default_max_transition을 그대로 사용해 현실적인 DRC 기준 확보.

6.5 ns vs 6.4 ns — 수렴 / 미수렴 대조

타이밍은 양쪽 모두 만족 — 차이는 fanout 위반 해소 여부.

항목	6.5 ns (수렴)	6.4 ns (미수렴)
CLK critical / slack	6.394 ns / +0.000066	6.293 ns / +0.001537
CLK 위반 paths	0	0
SRAM_CLK critical / slack	3.198 ns / +0.000751	3.145 ns / +0.000704
SRAM_CLK 위반 paths	0	0
max_fanout 위반	0	1 (DU/alu_unit/n569, 11 > 10)
Leaf cell / Buf·Inv cell	1,643 / 381	1,659 / 406
Design area	42,905.80 μm²	43,442.05 μm²
Total power	11.538 mW	11.487 mW

6.4 ns는 타이밍을 오히려 여유 있게 닫았지만, buffer를 더 투입(406 vs 381)하고도 fanout 위반 1개를 없애지 못함 — 180 nm의 실질적 하한은 6.5 ns.

06

SECTION

TSMC 130 nm

topographical 입력 파일 부재로 WLM 사용 — 2.6 ns 수렴 vs 2.5 ns 미수렴.

Using WLM(Wire-Load Model)

130 nm은 topographical에 필요한 tech DB가 없어 배선을 모델로 추정.

부재한 tech DB

tluplus(RC 정보) · mapping 파일,
metal 이름과 layer 정보가 담긴
tech 파일 등이 없음.
→ placement 기반 RC 계산 불가.

WLM의 추정 방식

fanout과 wire-load table만으로
net 길이와 부하를 추정.
실제 배치 정보는 사용하지 않음.

결과 해석

130 nm의 area · RC · timing은 근사치.
tech DB를 가진 32 nm과 180 nm이
서로 like-for-like 짝.

제약은 주기 2.6 ns · max_transition은 tsmc 130 lib의 default_max_transition · max_fanout은 8 그대로(180 nm과 달리 상향하지 않음).

Reference: Design Compiler · WLM · 정태웅 · 청주대학교

2.6 ns vs 2.5 ns — 수렴 / 미수렴 대조

CLK은 양쪽 다 닫혔지만 SRAM_CLK half-cycle 경로에서 갈림.

항목	2.6 ns (수렴)	2.5 ns (미수렴)
CLK critical / slack	2.438 ns / +0.000051	2.337 ns / +0.000599
CLK 위반 paths	0	0
SRAM_CLK critical / slack	1.255 ns / +0.000230	1.211 ns / −0.001207
SRAM_CLK 위반 paths / TNS	0 / 0.000	2 / −0.002414
max_fanout 위반	0	1 (DU/alu_unit/n498, 10 > 8)
Leaf cell / Buf·Inv cell	1,464 / 276	1,612 / 367
Design area	23,636.41 μm²	26,729.08 μm²
Total power	2.317 mW	2.958 mW

2.5 ns는 셀을 크게 늘리고 area가 13 % 증가했는데도 SRAM_CLK setup 2 paths와 fanout 1개가 남음 — 130 nm(WLM)의 하한은 2.6 ns.

07

SECTION

Comparison & Wrap-Up

세 공정의 결과를 나란히 두고 공정 · topo/WLM 차이를 정리.

정태웅 · 청주대학교

3-Way 비교 — 공정별 최종

각 공정에서 수렴한 주기 기준 결과. 동일 RTL · SRAM 매크로, 주기와 DRC 제약은 상이.

항목	SAED32 32 nm	TSMC 180 nm	TSMC 130 nm
수렴 주기 / 미수렴 주기	2.0 ns / 1.7 ns	6.5 ns / 6.4 ns	2.6 ns / 2.5 ns
CLK critical path	1.890 ns	6.394 ns	2.438 ns
SRAM_CLK critical	0.959 ns	3.198 ns	1.255 ns
Design area	12,924.83 μm²	42,905.80 μm²	23,636.41 μm²
Macro (SRAM) area	9,003.13 μm²	9,003.13 μm²	9,003.13 μm²
Leaf cell / Buf·Inv	1,569 / 435	1,643 / 381	1,464 / 276
Total power	2.057 mW	11.538 mW	2.317 mW
합성 모드	Topographical	Topographical	WLM
max_transition / max_fanout	0.100 ns / 8	lib default / 10	lib default / 8
Formality	PASS · 119 points	PASS · 162 points	PASS · 276 points

공유 — 동일 RTL · flow · SAED32 SRAM1RW64x8 매크로(표준셀만 공정별로 교체).

공정별 차이 해석

같은 RTL과 같은 fix를 적용해도 노드의 물리가 결과의 상한을 정함.

Feature size

공정이 커지면 셀도 커짐.
area 12,925(32 nm) → 23,636(130 nm)
→ 42,906 μm²(180 nm).

Transistor speed

미세 공정일수록 스위칭이 빠름.
수렴 주기 2.0 / 2.6 / 6.5 ns로 노드가 클수록 주기가 길어짐.

Capacitance

큰 공정일수록 capacitance가 큼.

Power & layout

게이트당 에너지가 커져 180 nm은 11.5 mW.
32 nm은 총 2.06 mW지만 그중 leakage가 1.25 mW.

SRAM macro 9,003 μm²는 전 공정 동일 — 32 nm 3,922 · 130 nm 14,633 · 180 nm 33,903 μm². 같은 설계와 같은 fix를 써도 공정이 area · power · timing의 한계를 정함.

Topographical vs WLM

130 nm만 WLM이라 비교를 읽는 기준 자체가 달라짐.

Topographical (32 · 180 nm)

tech DB를 이용해 rc값을 추정.
실측에 근접한 값.

WLM (130 nm)

Fanout 기반 rc값 추정.
실제 물리 정보를 반영하지 않아 값의 편차가 생길 수 있음.

비교 기준

32 nm ↔ 180 nm은 직접 비교 가능.
130 nm 수치는 참고치로 해석.

동일 조건에서 비교하려면 130 nm에도 tech · tluplus · map file 등 물리 정보가 필요함.

총정리

RTL 검증부터 세 공정 합성과 speed 한계 탐색까지의 결과 요약.

검증

RTL VCS로 done/pass를 확인하고
전 공정 Formality PASS.
SAED32는 gate-level VCS까지 재현.

한계 탐색

SAED32 1.8 ns / TSMC 180 6.5 ns /
TSMC 130 2.6 ns가 각 하한.
한 단계 더 줄이면 모두 위반이 남음.

자동화

split_fanout · report_transition_drivers로
fanout · transition 위반을 자동 처리 —
1.8 ns 이후 전 공정에 활용.

미수렴한 것들은 공통적으로 셀과 buffer를 더 투입하고 area · power가 늘었는데도 위반이 남음 — 같은 설계와 같은 fix로는 더 밀어낼 수 없는 지점임.